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Bachelor's Thesis

Perceived School-Related Stress and Academic Achievement
across School Tracks in German Secondary Education:
The Role of Contextual Stressors and Resources

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1. Introduction

School is one of the major developmental contexts of adolescence. It structures a substantial part of young people's everyday lives and provides an institutional setting for learning, performance evaluation, and social development ([Eccles & Roeser, 2011](#)). The extent of this institutional presence is illustrated by the German School Barometer, which reports that children and adolescents spend approximately 8,000 hours of their lives in school ([Robert Bosch Stiftung, 2024](#)). Yet the significance of schooling extends beyond the time students spend there. The educational stressors hypothesis argues that educational expansion¹ and the growing importance of educational performance for later life chances may expose adolescents to increasing school-related pressure ([West & Sweeting, 2003](#); [Högberg, 2021](#)).

Young people therefore encounter a range of potential stressors within school, with academic demands representing a particularly common source of pressure ([García-Moya et al., 2023](#); [Huang et al., 2026](#)). Across 32 countries, perceived school pressure tended to increased among girls between 2002 and 2018, whereas trends among boys were inconsistent ([Löfstedt et al., 2020](#)). A systematic review by Wuthrich et al. ([2020](#)) provides a broader indication of this burden: across studies from 17 countries, approximately one in six students in the final years of secondary school experienced excessive distress.² Similar concerns are evident in contemporary Germany. In the 2025/26 German School Barometer, 61% of students reported that much was expected of them at school, while 47% indicated that they needed to study on weekends to meet school demands ([Robert Bosch Stiftung, 2026](#)).

These figures illustrate the relevance of school-related burden, but exposure to academic demands does not imply that all students experience the same degree of stress. Similar expectations, examinations, workloads, or school situations may be manageable for some students and highly burdensome for others. The present study therefore focuses on *perceived school-related stress*, referring to students' subjective appraisal of school-related demands and situations as burdensome rather than to the objective presence of

¹ Educational expansion refers to the long-term increase in participation in and attainment of higher levels of education ([Meyer et al., 1992](#); [Schofer & Meyer, 2005](#)). In the educational stressors perspective, this expansion is also linked to the growing importance of educational performance and qualifications for later life chances ([Högberg, 2021](#)).

² Following Hans Selye's distinction, *distress* refers to negatively experienced or detrimental stress, whereas *eustress* denotes a more positive or adaptive form of stress ([Selye, 1974](#); see also [chapter 2.1](#)).

particular demands.³ This distinction is central to the study because students encounter school within different institutional, family, social, and personal conditions that may shape how demands are experienced and how students respond to them. The consequences of perceived school-related stress may, however, extend beyond students' immediate stress experiences to their academic development. This raises the question of whether students reporting greater school-related stress subsequently perform less well at school. This concern is also reflected in current German media coverage, where school-related pressure and anxiety are discussed not only as threats to students' well-being, but also as potential barriers to learning and academic performance ([Schiller, 2025](#); [Scholz, 2026](#)).

State of Research

Despite considerable variation in how academic and school-related stress is conceptualized and measured ([Putwain, 2007](#); [Pascoe et al., 2020](#)), longitudinal studies generally point to a negative relationship with later academic achievement. Kaplan et al. ([2005](#)) followed 1,034 adolescents from junior high school into high school and found that school-related stress predicted poorer academic performance three years later. Schraml et al. ([2012](#)) similarly reported poorer final grades among Swedish students with persistent severe stress symptoms.

German longitudinal research points in the same direction. Grützmacher and Raufelder ([2019](#)) followed students from early to middle adolescence and found that earlier perceived stress negatively predicted subsequent school achievement. More recently, Becker and Börnert-Ringleb ([2026](#)) examined 644 German secondary-school students across three measurement points. Higher school-related stress was associated with poorer later achievement, whereas the reverse relationship from achievement to later stress was less consistent. Their study is particularly relevant to the present research because it combines a school-related stress measure with longitudinal data from German secondary education. Achievement, however, was assessed through a global teacher rating rather than subject-specific grades.

Liu and Lu ([2011](#)), however, found that the negative stress-achievement association was not uniform across students. This raises the question of whether the relationship between stress and achievement depends on the conditions in which students experience

³ Research on stress frequently uses terms such as academic pressure, stressors, perceived stress, and stress responses for related but conceptually distinct phenomena ([Epel et al., 2018](#); [Becker & Börnert-Ringleb, 2026](#)).

and manage school demands. Vogelaar et al. (2024), for example, found higher stress concerning school performance and school-leisure conflict among students in academic rather than vocational education, despite similar overall stress levels across tracks. German evidence further indicates that contextual processes may differ across school settings. Kulakow et al. (2021) found that parental pressure and support were associated with stress development differently across high- and low-track schools. These findings suggest that school-related stress and its consequences may be embedded in both institutional settings and broader conditions surrounding students' everyday lives.

Taken together, longitudinal evidence links school-related stress to later academic achievement, including in German secondary education, but considerably less is known about whether this relationship differs across German secondary-school tracks, how it is connected to students' family, school, social, and personal conditions, and which of these conditions are associated with later stress beyond earlier stress levels.

The present study addresses these questions using longitudinal data from the German TwinLife study. It first examines whether perceived school-related stress is associated with later German and mathematics grades after earlier subject-specific achievement is taken into account. Second, it tests whether this relationship differs across academic, non-academic, and comprehensive school tracks. Third, it examines how the stress-achievement relationship is connected to family, school, social, and personal conditions. Finally, the study explores whether these contextual conditions are associated with later school-related stress beyond students' earlier stress levels and whether these relationships differ across school tracks. By linking students' subjective stress experiences to school-track membership and to family, school, social, and personal conditions, the study connects longitudinal research on school-related stress with a social-scientific perspective on the institutional and social contexts of educational development.

The following chapters develop the theoretical framework and derive the hypotheses and research questions, describe the data and analytical strategy, present the empirical results, and discuss their interpretation, limitations, and implications.

2. Theoretical Background and Hypotheses

Building on the research gaps identified in the introduction, this chapter develops the theoretical basis of the study. It first clarifies the concept of perceived school-related stress and the German secondary-school context, then integrates the relevant theoretical perspectives and derives the hypotheses and research questions.

2.1. Conceptualizing Stress

The term stress is ubiquitous in both scientific and everyday discourse. Its prominence has even led some authors, somewhat rhetorically, to describe the present century as a “stress era” ([Shahsavarani et al., 2015](#): 230). Yet its widespread use has not produced corresponding conceptual clarity ([Fomina et al., 2025](#): 89; [Putwain, 2007](#)). As Hans Selye, a central figure in the development of modern stress research, famously observed, “stress, like relativity, is a scientific concept which has suffered from the mixed blessing of being too well known and too little understood” ([1980](#): 127).

Selye’s work advanced a primarily physiological, response-based understanding of stress, conceptualizing it as a nonspecific bodily response to demands ([Selye, 1976](#)). This perspective became foundational for stress research, but offers limited leverage for explaining why similar demands may be experienced very differently across individuals. Lazarus and Folkman ([1984](#)) provide the central conceptual shift for the present study by locating psychological stress in the relationship between the person and the environment. Stress arises when this relationship is appraised as taxing or exceeding available resources and threatening well-being ([Lazarus & Folkman, 1984](#): 21). External demands are therefore not inherently stressful; their relevance depends on how they are evaluated in relation to available coping resources.

Contemporary stress research further differentiates the stress process by separating stressor *exposure*, subjective perceptions and *appraisals*, and subsequent psychological or physiological *responses* more explicitly ([Epel et al., 2018](#); [Crosswell & Lockwood, 2020](#)). Keeping these components analytically separate avoids conflating the conditions that may give rise to stress with their subjective evaluation and subsequent responses.⁴

Within educational research, perceived academic stress has similarly been conceptualized as the experience of academic demands as taxing available resources ([Huang et al., 2026](#)). Building on this appraisal-oriented logic, the present study understands perceived school-related stress more broadly as students’ subjective experience of school-related demands and experiences as burdensome or difficult to manage. Accordingly, perceived

⁴ The appraisal-based perspective adopted here represents one strand within the wider stress literature. Other approaches emphasize objective stressor exposure or physiological responses, including biomarkers such as cortisol or alpha-amylase ([Hoferichter et al., 2024](#); [Ringeisen et al., 2019](#)). Stress has also been studied across general, occupational, and higher-education contexts (e.g., [Gross & Seebaß, 2014](#); [Akgun & Ciarrochi, 2003](#); [Kötter et al., 2017](#); [Ringeisen et al., 2019](#)). These approaches address different components and settings of the stress process.

school-related stress is treated separately from the contextual conditions that may contribute to it, which are developed as potential stressors and resources in [Section 2.3](#).

2.2. School Tracking in German Secondary Education

The German secondary-school system provides the institutional context for the present study. Because responsibility for school education lies primarily with the 16 Länder, secondary-school structures vary across Germany ([KMK, 2025](#)). Following primary education, which generally covers Grades 1 to 4 and extends through Grade 6 in Berlin and Brandenburg, students enter lower secondary education, where schooling is differentiated into three principal educational tracks: Hauptschule, Realschule, and Gymnasium. Depending on the Land, these tracks may be offered in separate school types or combined within schools providing two or three educational programmes ([KMK, 2025](#)).

The German system can therefore no longer be adequately described as a uniformly tripartite structure of Hauptschule, Realschule, and Gymnasium. Over recent decades, several Länder have merged lower and intermediate school types, introduced schools combining multiple educational programmes, or expanded comprehensive structures, while the Gymnasium has generally remained institutionally distinct ([Becker et al., 2017](#); [KMK, 2026](#)). The present study does not attempt to reproduce every Länder-specific school structure. Instead, the observed school types are grouped into three analytical categories: academic track, non-academic track, and comprehensive school. In the terminology used throughout this study, the academic track refers to Gymnasium-oriented schooling, whereas the non-academic category combines lower and intermediate general secondary-school types and should not be equated with vocational education. Comprehensive schools refer to integrated secondary-school settings that combine multiple educational pathways within the same school.

Despite this structural diversity, a central characteristic of German secondary education remains the comparatively early differentiation of students into distinct educational environments. Between-school tracking is intended to reduce achievement heterogeneity within learning groups and to allow curricula, instructional pace, and educational goals to be adapted to students' prior achievement and educational pathways. However, track placement is not determined by achievement alone. Transition rules differ between the Länder, including the relative influence of teacher recommendations and parental choice ([KMK, 2025](#)), and research consistently shows that social background is associated with track placement even beyond differences in prior achievement ([Maaz et al., 2008](#); [Grewenig, 2021](#); [Herrmann & Bach, 2025](#)).

This selectivity is important for understanding school tracks as *differential learning environments*. Differences between tracks reflect both *compositional* and *institutional* processes ([Baumert et al., 2006](#); [Maaz et al., 2008](#)). Compositional differences arise because student populations vary in characteristics such as prior achievement and socioeconomic background. Institutional differences concern the learning environment itself, including curricula, instructional practices, teacher qualifications, educational expectations, and learning opportunities. Both processes are closely intertwined, as selective transitions create different student compositions while subsequent exposure to different educational environments can contribute to diverging educational development.

German longitudinal evidence is consistent with this interpretation. Traini et al. ([2021](#)) found stronger competence growth in the academic track even among students with similar initial conditions, with class-level achievement composition accounting for part of the difference. Schiepe-Tiska ([2019](#)) additionally documented track differences in teaching quality and in its associations with mathematics competence, interest, and self-efficacy. Using longitudinal competence data measured before and after secondary-school tracking, Herrmann and Bach ([2025](#)) likewise found weaker mathematics competence development among initially comparable students attending non-academic rather than academic school tracks.

For the present study, school-track membership is therefore treated as an indicator of the broader institutional context in which students' achievement and school experiences are embedded.⁵

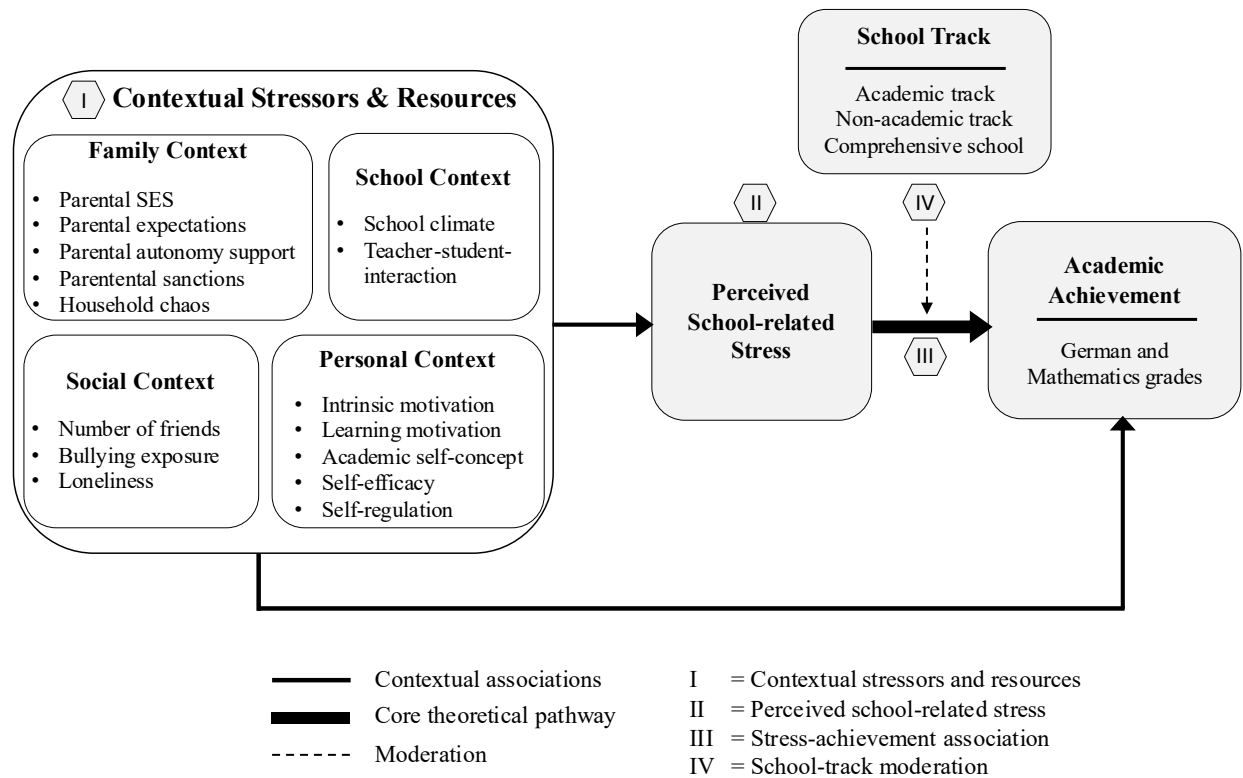
2.3. Integrative Theoretical Framework

The following framework integrates four related theoretical issues that underpin the hypotheses and research questions formulated below. These concern how school demands become subjectively stressful, how these experiences are embedded in students' broader environments and personal resources, how perceived stress may be linked to academic achievement, and why this association may differ across school tracks. No single theoretical perspective accounts for all of these relationships. The framework therefore combines complementary approaches, each serving a specific role.

⁵ It should be noted that this variable does not capture specific features of schooling such as curricula, instructional practices, or classroom composition in a direct way. Instead, it serves as a proxy for systematically different educational environments, which provides the theoretical basis for examining whether the relationship between perceived school-related stress and later achievement varies across the three analytical groups. The corresponding moderation argument is developed in [Section 2.3](#) and translated into H2 in [Section 2.4](#).

Its general architecture builds on the integrative stress and coping framework proposed by Moos and Holahan (2003), which connects environmental and personal systems with stressful conditions, appraisal and coping processes, and subsequent adjustment. Morazes (2016) provides a related educational application by shifting the outcome focus towards academic success. Building on this structure, the present framework places perceived school-related stress at its centre, treats German and mathematics grades as educational outcomes, and incorporates school track as an institutional context. Figure 1 presents the empirically focused version of the framework, while Appendix Figure A1 retains theoretically relevant elements of the broader stress process that are not directly observed in the main analyses.

Figure 1. Integrative theoretical framework of perceived school-related stress, contextual stressors and resources, school track, and academic achievement



Note. The figure summarizes the theoretical and analytical relationships examined in the study. Arrows indicate hypothesized associations and organizing pathways and should not be interpreted as evidence of causal effects. Own representation based on and extending the frameworks of Moos and Holahan (2003) and Morazes (2016). The exploratory context-by-school-track interactions examined in RQ4 are *not* separately depicted.

Perceived school-related stress as an appraisal-based construct (Panel II)

Building on Section 2.1, Panel II distinguishes *perceived* school-related stress from specific school demands and contextual conditions. Similar examinations, workloads, parental expectations, or classroom situations may be experienced differently depending on how students evaluate these demands relative to their available resources (Lazarus &

[Folkman, 1984](#)). Perceived school-related stress therefore occupies the appraisal-related centre of the framework, reflecting students' subjective experience of school-related demands rather than the demands themselves. Contextual conditions are treated separately as potential stressors and resources (see Panel I). This distinction allows the framework to connect students' subjective stress experiences to their broader educational environments without equating those environments with stress itself.

Contextual stressors and resources (Panel I).

Panel I situates perceived school-related stress within family, school, social, and personal contexts. The theoretical basis for organizing these conditions as potential stressors and resources is Conservation of Resources theory (COR; [Hobfoll, 1989](#)). Hobfoll conceptualizes *resources* broadly as objects, personal characteristics, conditions, and energies that are valued in themselves or facilitate the attainment of valued outcomes. Stress arises particularly when valued resources are threatened or lost, or when substantial resource investment fails to produce the expected gain ([Hobfoll, 1989](#): 516).

Applied to schooling, contextual conditions are not inherently protective or stressful. They may provide resources for handling school demands, consume such resources, or create additional demands. The four contextual domains in [Figure 1](#) therefore represent different conditions surrounding students' stress experiences and academic development.⁶

Family Context. The family can shape both the resources students have available for dealing with school demands and the pressures they experience around achievement. *Parental socioeconomic status (SES)* is included because socioeconomic background is closely associated with educational opportunities and academic achievement ([Sirin, 2005](#)). *Parental autonomy support* represents a potentially protective family condition, as autonomy-supportive parenting has been associated with more adaptive school adjustment and achievement ([Joussemet et al., 2005](#)). By contrast, *parental sanctions* following poor grades capture potentially pressuring parental responses to academic performance, while *household chaos* reflects instability and disorganization within the home environment. Household chaos, in particular, has been consistently associated with adverse cognitive and academic outcomes across previous research ([Marsh et al., 2020](#)).

Parental expectations are less straightforward. Expectations can communicate guidance and confidence in students' abilities, but they may also become a source of academic

⁶ The four contextual domains are theoretically broader than their empirical operationalization. The specific indicators used in the analyses are constrained by the measures available in TwinLife at the relevant observation wave and are therefore not intended to provide an exhaustive representation of each domain; see [Section 3.2](#) for their operationalization.

pressure when students experience them as difficult to fulfil. Contemporary academic-stress research explicitly identifies expectations from parents (and teachers) as one source of perceived academic stress, while longitudinal evidence also shows that parental pressure and support are differently associated with adolescents' stress development ([Huang et al., 2026](#); [Kulakow et al., 2021](#)).

School Context. The school environment can shape how manageable academic demands appear to students. *Teacher-student interaction* captures students' experiences of fair treatment, teachers' willingness to listen, and additional help, while the *school-climate* measure reflects a relational and well-being-oriented aspect of the broader school environment. More positive student-teacher interactions have been associated with lower academic stress ([Banks & Smyth, 2015](#)). Perceived teacher support has also been linked to greater coping ability and lower helplessness when students face demanding situations ([Hoferichter et al., 2022](#)). García-Moya et al. ([2023](#)) further show that school-performance stress is related to classroom goal structures and teacher support.

Social Context. School is not only an academic setting but also an important social arena in which adolescents form friendships, compare themselves with others, and experience inclusion or exclusion ([Östberg et al., 2015](#)). Peer relations can therefore provide support when school demands are difficult to manage, but they can also become a source of stress themselves ([ibid.](#)). The present study captures different aspects of this social environment through number of friends, loneliness, and bullying exposure. *Number of friends* indicates the size of a student's social network, but does not show whether these relationships actually provide emotional or practical support. Network size and perceived support capture different dimensions of social relationships ([Cohen & Wills, 1985](#)). *Loneliness*, in contrast, captures perceived insufficiency in students' social connections, whereas *bullying exposure* captures reported adverse peer experiences.

Previous research provides direct support for considering these peer conditions in relation to school stress. Murberg and Bru ([2004](#)) identified difficulties with peers as one of four distinct dimensions of school-related stress among Norwegian adolescents. Banks and Smyth ([2015](#)) likewise found substantially higher stress among students reporting negative peer relations, particularly among those exposed to bullying. More specifically, Östberg et al. ([2018](#)) conceptualized bullying as a repeated social-evaluative stressor and found bullied adolescents to report higher perceived stress and recurrent pain. Conversely, supportive peer relations may function as resources. Hoferichter et al. ([2022](#)) found that

classes characterized by stronger classmate support showed higher coping ability and academic achievement.

Personal Context. Students differ not only in the demands and support they encounter, but also in the personal resources they bring to school. COR theory allows such personal characteristics to be understood as resources when they help individuals manage demands or protect valued outcomes ([Hobfoll, 1989](#)). The present study considers academic self-concept, self-efficacy, intrinsic motivation, learning motivation, and self-regulation as different aspects of these personal academic resources.

Academic self-concept concerns students' perceptions of their own academic abilities ([Shavelson et al., 1976](#)) and is closely intertwined with achievement: longitudinal research indicates reciprocal relations in which prior achievement shapes subsequent academic self-perceptions, which can in turn contribute to later achievement ([Marsh & Martin, 2011](#)). *Self-efficacy* refers more specifically to students' confidence that they can successfully handle particular demands (e.g. [Bandura, 1997](#)). This perceived capability is closely related to stress appraisal. Ringeisen et al. ([2019](#)) found that stronger self-efficacy was associated with lower threat appraisal and lower anxiety around an examination, indicating that perceived capability is related to how demanding an achievement situation is experienced.

Intrinsic motivation and *learning motivation* capture students' willingness to engage with learning, distinguishing motivational orientation from perceived capability. Motivation is relevant to the present framework because it influences sustained engagement with academic tasks and is itself closely connected to stress. In a three-year study of Chinese high-school students, Liu ([2015](#)) found that earlier academic stress predicted lower subsequent intrinsic motivation and higher amotivation, illustrating that stress and students' willingness to engage academically can develop together over time. Finally, *self-regulation* concerns students' capacity to organize and control their own goal-directed behavior. Self-regulated learning includes processes such as goal setting, planning, monitoring progress, and adjusting behavior when difficulties arise ([Zimmerman, 2002](#)), while more recent educational stress research identifies self-regulation as a resource for maintaining performance under stressful academic conditions ([Fomina et al., 2025](#)).

These factors are therefore included because they capture different personal conditions that may affect both how students deal with school demands and how successfully they learn and perform. Their close connection to academic achievement also means that

they cannot be treated simply as external background characteristics, a point that becomes important when interpreting the context-adjusted models.

From perceived school-related stress to academic achievement (Panel III)

The appraisal-based conception of stress introduced in [Section 2.1](#) explains why school demands may become subjectively stressful, but not how this experience may subsequently interfere with learning and academic performance. Panel III therefore draws on Pekrun's control-value theory of achievement emotions (CVT; [Pekrun, 2006](#)) to specify processes that can connect stressful school experiences with achievement.

CVT proposes that achievement emotions arise particularly from students' appraisals of *control* over achievement activities and outcomes and the *value* they attach to them. When important academic outcomes are perceived as difficult or uncertain to control, negative achievement emotions such as anxiety or hopelessness become more likely ([Pekrun, 2006](#)). This links CVT to the appraisal-based stress perspective developed above. Lazarus and Folkman focus on whether demands are experienced as exceeding available resources, whereas CVT focuses more specifically on perceived control and value in achievement situations. Achievement emotions therefore provide one possible link between perceived school-related stress and later achievement, while remaining conceptually distinct from school-related stress itself.⁷

Achievement emotions can affect learning and performance through several pathways. They can direct cognitive resources away from the task, influence students' intrinsic and extrinsic motivation, affect the learning strategies they use, and support or impede self-regulated learning ([Pekrun, 2006](#)). Anxiety illustrates that these processes are not uniformly negative: worries about possible failure can occupy task-relevant cognitive resources and undermine intrinsic motivation, while anxiety may simultaneously motivate additional effort to avoid failure ([ibid.](#)). The overall effect on achievement therefore depends on how these different cognitive and motivational processes combine.

Applied to school-related stress, students who experience school demands as difficult to manage may consequently face emotional, cognitive, and motivational conditions that make effective learning and performance more difficult. Consistent with this theoretical connection, Becker and Börnert-Ringleb ([2025](#)) found school-related stress and

⁷ CVT is used here as a complement to the appraisal-based stress perspective rather than as a competing explanation. Pekrun ([2024](#)) explicitly identifies Lazarus' transactional model of stress and emotions as one of the appraisal approaches on which the development of CVT built.

achievement anxiety to be strongly associated among German secondary-school students while treating them as distinct constructs.⁸

School tracks as institutional moderators (Panel IV)

Students who experience school as stressful still have to attend classes, prepare for assessments, complete tasks, and maintain performance. Whether perceived stress is more strongly associated with poorer achievement may therefore depend on the educational environment in which students continue to meet these demands. Stage-environment fit theory (SEF; [Eccles et al., 1993](#)) provides a useful perspective on this issue. It proposes that adolescents' academic and motivational development is more favorable when the demands and opportunities of their educational environment correspond to their needs and capacities, whereas mismatches can undermine motivation, self-perceptions, and academic functioning.

This fit perspective complements the resource argument developed in Panel I. If perceived school-related stress indicates that students already experience school demands as taxing their available capacities, the conditions under which they continue learning become particularly relevant. Support, manageable instructional demands, and effective learning opportunities may help students maintain engagement and performance despite experiencing stress. Conversely, additional demands or insufficient resources may make it harder to compensate for the cognitive and motivational difficulties outlined in Panel III. School track enters the framework at this point. It is used as a broad institutional indicator of the educational environment rather than as a direct measure of stage-environment fit itself. Tracks can differ in the demands students encounter and in the resources and learning opportunities available to them, as established in [Section 2.2](#).

German evidence shows that relationships between educational conditions and student outcomes can differ across school tracks. Schiepe-Tiska ([2019](#)), for example, found that associations between teaching quality and mathematics competence, interest, and self-efficacy varied across school tracks. This does not establish a stress-by-track effect, but illustrates that associations between educational conditions and student outcomes can vary across track environments.

⁸ The present study does not directly measure achievement emotions or the cognitive and motivational processes described by CVT. It therefore cannot determine whether anxiety, reduced cognitive resources, changes in motivation, or learning strategies mediate the association between perceived school-related stress and later grades.

2.4. Hypotheses and Research Questions

The framework developed above leads to two hypotheses concerning the stress-achievement association and two research questions addressing contextual conditions and later school-related stress.

Longitudinal studies reviewed above generally show that higher school-related stress is associated with poorer later academic outcomes, including in German secondary education ([Kaplan et al., 2005](#); [Grützmacher & Raufelder, 2019](#); [Becker & Börnert-Ringleb, 2026](#)). Panel III further suggests that emotional, cognitive, and motivational processes may contribute to this relationship ([Pekrun, 2006](#)). This leads to the first hypothesis:

H1: Higher perceived school-related stress is associated with lower subsequent German and mathematics grades.

Panel IV extends this relationship to the educational environments described in [Section 2.2](#). German school tracks differ in student composition, instructional conditions, and learning opportunities ([Schiepe-Tiska, 2019](#); [Traini et al., 2021](#); [Herrmann & Bach, 2025](#)). From a stage-environment fit perspective, such differences may alter the conditions under which students have to maintain academic performance while experiencing stress. However, the framework does not provide a clear basis for predicting in which track the negative association should be strongest. The second hypothesis is therefore non-directional:

H2: The association between perceived school-related stress and subsequent achievement differs across school tracks.

Panel I situates perceived school-related stress within students' family, school, social, and personal conditions. Students reporting higher stress may simultaneously differ in the support, demands, and personal resources they experience, while some of these conditions are themselves associated with later achievement. The observed stress-achievement association may therefore change when these conditions are taken into account. This motivates the third research question:

RQ3: To what extent does the association between perceived school-related stress and subsequent German and mathematics grades remain after accounting for family, school, social, and personal conditions?

The final research question shifts the focus from the *consequences* of school-related stress to its *development* over time. It examines whether contextual conditions are associated with later stress beyond students' already existing differences in stress.⁹ Panel I provides a resource-based rationale for linking these contextual conditions to students' stress experiences, but it does not specify how their associations with later stress should differ across school tracks. The track-specific part of RQ4 is therefore motivated primarily by empirical findings from related research. Kulakow et al. (2021) found that parental pressure and support were related to stress development differently in high- and low-track schools, while Vogelaar et al. (2024) found higher stress concerning school performance and school-leisure conflict among students in academic than vocational education. Beyond stress research, Bittmann (2024) similarly showed that parental and peer associations with students' aspirations differed between academic and non-academic tracks. These findings provide a basis for examining track differences, but do not support a directional expectation across the full set of contextual conditions. RQ4 is therefore formulated exploratively:

RQ4: Which family, school, social, and personal conditions are associated with later perceived school-related stress, after accounting for earlier stress, and do these associations differ across school tracks?

3. Data and Methods

This chapter translates the theoretical framework into the empirical design. It first introduces the TwinLife data and longitudinal study design, then describes the operationalization of the study variables, and the contextual conditions, and finally specifies the statistical strategy used to evaluate H1 to H2 and address RQ3 and RQ4.

3.1. Data and Analytical Design

This study uses data from TwinLife, a longitudinal twin-family study following twins, siblings, and their families across childhood, adolescence, and early adulthood in Germany (Mönkediek et al., 2019). The analyses use TwinLife Version 9.0.0 (Diewald et al., 2025). TwinLife includes five face-to-face (F2F) survey waves, denoted here chronologically as T0 to T4 (F2F1 to F2F5).

⁹ In RQ3, contextual conditions enter as additional covariates in models relating school-related stress to later achievement. In RQ4, these same conditions become focal predictors of later school-related stress, while earlier stress is included as a control.

The main achievement analyses follow a three-wave lagged dependent variable (LDV) design ([Keele & Kelly, 2006](#)). Prior German and mathematics grades are measured at T0 (F2F1, 2014–2015), perceived school-related stress and contextual conditions at T1 (F2F2, 2016–2017), and subsequent German and mathematics grades at T2 (F2F3, 2018–2019). This ordering allows the stress-achievement association to be examined while accounting for prior achievement in the same subject.¹⁰ Twins and siblings are retained as separate student observations, with within-family dependence addressed through family-clustered standard errors ([Section 3.3](#)).

After applying the three-wave and school-track criteria, the analytical samples comprise 845 students for German and 843 for mathematics. The estimation samples become smaller when additional covariates are included because complete information is required for all variables in the respective model specification. In particular, the sequential M1–M4 achievement models are estimated on a common complete-case sample to ensure that changes across models cannot result from changes in sample composition. The resulting model-specific sample sizes are reported with the respective analyses.

RQ4 uses a different longitudinal specification because the outcome is later school-related stress rather than achievement. Contextual conditions measured at T1 are related to school-related stress at T2 while school-related stress at T1 is included as the prior outcome measure. RQ4 additionally examines whether these contextual associations differ across school tracks.

3.2. Measures and Operationalization

[Table 1](#) reports descriptive statistics for the variables used in the main analyses. The descriptions below focus on their operationalization; detailed item wording, response scales, index composition, and scale diagnostics are provided in [Appendix B](#). All multi-item indices were checked for internal consistency before their final construction. For scales with three or more items, Cronbach's α , McDonald's ω , and item-level diagnostics were inspected, while two-item indices were assessed using the correlation between their two items. These checks led to revised item compositions for parental expectations, household chaos, self-regulation, and bullying exposure ([Appendix Tables B1 and B2](#)). Unless noted otherwise, higher index values indicate higher levels of the respective construct.

¹⁰ The temporal separation of predictor and outcome and adjustment for prior achievement strengthen temporal ordering relative to a cross-sectional specification, but the LDV design does not by itself identify causal effects ([Keele & Kelly, 2006](#)).

Academic achievement. Academic achievement is measured using report-card grades in German and mathematics (Instinske et al., 2026). Grades are reverse-coded from the German grading scale so that higher values indicate better achievement, with 6 representing the highest and 1 the lowest achievement level. Both subjects are analyzed separately. In the three-wave school-track samples, later grades range from 2 to 6 and average approximately 4.3 in both subjects (Table 1).

Table 1. Descriptive Statistics of Variables in Main Analyses

Variable	Range	German			Math		
		M / %	SD	n	M / %	SD	n
Prior grade	2-6	4.51	0.81	845	4.57	0.93	843
Later grade	2-6	4.28	0.89	845	4.26	1.02	843
School-related stress	1-4	1.89	0.52	845	1.89	0.52	843
School-track							
Academic track (Reference)	0/1	62.25	-	526	62.16	-	524
Non-academic track	0/1	19.53	-	165	19.57	-	165
Comprehensive school	0/1	18.22	-	154	18.27	-	154
Controls							
Age	13-20	14.08	2.02	845	14.08	2.01	843
Sex							
Female (Reference)	0/1	49.23	-	416	48.99	-	413
Male	0/1	50.77	-	429	51.01	-	430
Cohort							
Cohort 1	0/1	2.25	-	19	2.25	-	19
Cohort 2	0/1	83.55	-	706	83.63	-	705
Cohort 3	0/1	13.73	-	116	13.64	-	115
Cohort 4	0/1	0.47	-	4	0.47	-	4
Family Context							
Parental SES	15-89	62.35	17.66	810	62.35	17.61	808
Parental expectations	1-5	4.00	0.81	841	4.00	0.81	839
Parental autonomy support	1-5	4.28	0.74	799	4.28	0.74	797
Parental sanctions	1-5	2.39	1.03	837	2.39	1.04	835
Household chaos	1-5	2.23	0.84	793	2.22	0.84	791
School Context							
School climate	1-4	3.08	0.56	840	3.08	0.56	838
Teacher-student interaction	1-4	3.33	0.52	837	3.33	0.52	835
Social Context							
Number of friends	0-109	8.07	7.41	842	8.01	7.27	840
Loneliness	1-4	1.60	0.87	797	1.60	0.86	796
Bullying exposure	1-4	1.25	0.46	801	1.25	0.46	799
Personal Context							
Academic self-concept	1-5*	3.58	0.72	845	3.64	0.88	843
Intrinsic motivation	1-5	3.24	0.94	845	3.45	1.06	843
Learning motivation	1.33-5	3.88	0.69	845	3.89	0.70	843
Self-efficacy	1-5	3.89	0.59	723	3.89	0.59	722
Self-regulation	1-5	2.81	0.75	690	2.81	0.75	687

Note. Descriptive statistics are based on the final school-track samples used in the main three-wave LDV design. Ranges refer to *empirically* observed values. M denotes the mean for continuous variables and % the percentage for categorical variables; n denotes the number of valid observations. Grades are reverse-coded. The academic track and female students serve as reference categories in the regression models. *For academic self-concept, the observed range differs between the subject-specific samples (German: 1.67–5; mathematics: 1–5).

Perceived school-related stress. Perceived school-related stress is measured with seven TwinLife items covering school-related tension, difficulties falling asleep, irritability, rumination about school problems, exhaustion, perceived school pressure, and insufficient

time for other activities (Appendix [Table B1](#)). The index is calculated as the mean of all seven items and ranges from 1 to 4. In the analytical samples, perceived school-related stress averages 1.89 (SD = 0.52; [Table 1](#)). Internal consistency is acceptable (Cronbach's $\alpha = .77$; McDonald's $\omega = .77$; Appendix [Table B2](#)) and remains similar across the three school-track groups ($\alpha = .75$ – $.78$; Appendix [Table B3](#)). Exploratory factor analysis indicates that a two-factor representation fits the seven items better than a one-factor solution, although the two factors are strongly correlated ($r = .70$; Appendix [Table B4](#)). Given the strong correlation between the two factors and the study's broad conceptualization of perceived school-related stress, the seven-item mean index is retained for the main analyses.

School track. The TwinLife school-type information is recoded into three categories corresponding to the analytical distinction developed in [Section 2.2](#), namely academic track, non-academic track, and comprehensive school. School types that cannot be meaningfully assigned to these categories are excluded from the school-track comparison. The complete mapping of the original TwinLife school-type categories and the corresponding exclusions is reported in Appendix [Table B5](#).

Basic controls. Age, sex, and TwinLife birth cohort are included as basic individual characteristics. Age is measured in years, sex distinguishes male and female respondents, and cohort identifies the four TwinLife birth cohorts¹¹ represented in the analytical sample. Their distributions are reported in [Table 1](#).

Contextual conditions. The contextual variables are organized into the family, school, social, and personal contexts of students developed in [Section 2.3](#). With the exception of parental SES, all measures are based on students' own reports. The number of variables differs across contexts, with five capturing the family context, two the school context, three the social context, and five variables the personal context. This reflects the measures available for the longitudinal design rather than differences in their theoretical importance.

Family context. The family context includes parental socioeconomic status (SES), parental expectations, parental autonomy support, parental sanctions, and household chaos. Parental SES is operationalized as the highest available parental International Socio-Economic Index of Occupational Status (ISEI) score in the household.¹² Parental

¹¹ TwinLife defines its four birth cohorts based on the twins' birth years: 2009/2010 (Cohort 1), 2003/2004 (Cohort 2), 1997/1998 (Cohort 3), and 1990–1993 (Cohort 4; [Diewald et al., 2025](#)).

¹² The ISEI variable is derived from the TwinLife dataset and based on the International Socio-Economic Index of Occupational Status ([Ganzeboom et al., 1992](#)).

expectations and household chaos are based on two items each, while parental autonomy support and parental sanctions each comprise three items (Appendix [Table B1](#)).

School context. The school context includes school climate and teacher-student interaction. School climate combines two items capturing generalized perceptions of student-teacher relations and teachers' concern for students' well-being.¹³ Teacher-student interaction captures students' own experiences with the teachers who teach them through three items concerning fair treatment, additional help, and teachers' willingness to listen (Appendix [Table B1](#)).

Social context. The social context includes number of friends, loneliness, and bullying exposure. Number of friends is measured as a count variable and ranges from 0 to 109 in the three-wave school-track samples ([Table 1](#)). Loneliness is measured with a single item, while bullying exposure is a three-item index covering teasing, rumors, and social exclusion (Appendix [Table B1](#)).

Personal context. The personal context includes academic self-concept, intrinsic motivation, learning motivation, self-efficacy, and self-regulation. Academic self-concept and intrinsic motivation are available as both subject-specific and general-school measures. The German and mathematics achievement models use the corresponding subject-specific indices, whereas the general-school versions are used in the prospective models of later school-related stress, for RQ4. Academic self-concept, intrinsic motivation, learning motivation, and self-efficacy are each based on three items, while self-regulation comprises five items. The self-regulation items are formulated in the direction of lower self-regulation; the resulting index is therefore reversed so that higher scores indicate stronger self-regulation (Appendix [Table B1](#)).

3.3. Statistical Strategy

German and mathematics grades are modeled separately using ordinary least squares (OLS). OLS is retained as the main approach because the coefficients are directly interpretable in reverse-coded grade points. Since report-card grades are bounded and ordinal, the central models are additionally estimated using ordinal logistic regression to assess whether the conclusions depend on treating grades as approximately metric (Appendix [Table C7](#)). Family-clustered standard errors account for dependence between twins and siblings from the same family ([Cameron & Miller, 2015](#)). School-related stress and age

¹³ School climate is a multidimensional construct that can encompass relational, academic, safety, and institutional dimensions ([Wang & Degol, 2016](#)). The two TwinLife items used here capture *only* a relational and well-being-oriented component and are interpreted accordingly.

are mean-centered in the achievement models; parental SES is additionally centered when included. Centering school-related stress makes school-track main effects interpretable at the average level of stress.

The main achievement models build sequentially on one another. M1 estimates the association between school-related stress and later achievement while accounting for prior subject-specific achievement, age, sex, and cohort. School track is added in M2, followed by the interaction between school-related stress and school track in M3. M4 adds selected contextual covariates to examine whether the stress-achievement association and the moderation pattern remain similar after these additional conditions are considered. For each subject, M1–M4 are estimated on the same complete-case sample defined by the variables required for M4, so that coefficient changes across the model sequence cannot result from changes in sample composition.¹⁴

The academic track serves as the reference category in the moderation models. The school-related stress coefficient therefore represents the association between stress and later achievement within the academic track, while the interaction terms represent differences from this slope in the non-academic track and comprehensive schools. Because school track has three categories, moderation is evaluated primarily using joint Wald tests of whether both interaction terms are simultaneously equal to zero. Track-specific slopes and pairwise comparisons between them are reported in [Appendix D \(Hayes & Montoya, 2017\)](#).

RQ3 examines the family, school, social, and personal contexts separately. For each domain, a block-specific baseline model (M0-A to M0-D) includes school-related stress, prior achievement, school track, parental SES, and the basic controls.¹⁵ The corresponding adjusted model (M1-A to M1-D) adds the variables belonging to that contextual domain. Each baseline-adjusted pair is estimated on exactly the same complete-case sample. Changes in the school-related stress coefficient can therefore be attributed to the inclusion of the measured contextual conditions rather than to a changing estimation sample. The resulting stability or attenuation of the coefficient is used to assess how strongly the observed stress-achievement association overlaps with each contextual domain. Because

¹⁴ The common estimation sample is defined by complete information on all variables required for M4. Consequently, M1 to M3 are estimated on slightly fewer observations than would be available for these models alone. This restriction ensures that coefficient changes across M1 to M4 cannot result from changes in sample composition.

¹⁵ Parental SES is retained as a common background control across all context blocks because socioeconomic background is closely related to both school-track placement and academic achievement. Consequently, the family-context comparison captures the additional contribution of family-process variables *beyond* parental SES.

school-related stress and most contextual conditions are measured at the same wave, attenuation is not interpreted as evidence of mediation.¹⁶

For RQ4, later school-related stress becomes the outcome and earlier school-related stress is included as a control. Separate models are again estimated for the family, school, social, and personal contexts. Each model first estimates the prospective associations of the contextual conditions with later stress and then adds their interactions with school track. Continuous contextual predictors and both stress measures are standardized so that coefficients can be interpreted in standard-deviation units and compared more readily across measures with different original scales. School-track differences are assessed using joint Wald tests of the interaction terms. Main results are reported in [Tables 7](#) and [8](#), with additional results in [Appendix H](#).

All analyses use complete cases for the variables required by the respective model; multiple imputation is not applied. [Appendix C](#) reports diagnostics concerning family clustering, multicollinearity, residual behavior, heteroskedasticity, influential observations, and the ordinal-logistic robustness checks. Supplementary two-wave and later-wave cross-sectional analyses in [Appendices F](#) and [G](#) assess whether the main stress-achievement and moderation patterns are broadly consistent under alternative specifications. The substantive interpretation remains based on the three-wave LDV models. All significance tests are two-sided.

4. Results

This chapter first presents the unadjusted patterns of school-related stress and achievement and then reports the longitudinal analyses for the hypotheses and research questions.

4.1. Descriptive Patterns of School-Related Stress and Achievement

The pairwise correlations provide an initial description of the relationship between school-related stress and later achievement ([Appendix Figure C2](#)). Higher stress is associated with lower later grades in both subjects, with a somewhat stronger correlation for mathematics (German: $\rho = -0.18$; mathematics: $\rho = -0.24$). Prior and later grades are more strongly related (German: $\rho = 0.52$; mathematics: $\rho = 0.56$).

Track-specific descriptives are reported in [Table 2](#). Mean school-related stress differs

¹⁶ Because school-related stress and most contextual conditions are measured at the same wave, their temporal ordering cannot be established. Changes in the stress coefficient after contextual adjustment are therefore interpreted as empirical overlap rather than evidence of mediation ([Cole & Maxwell, 2003](#); [Maxwell & Cole, 2007](#)).

little across school tracks: academic-track students average 1.92, compared with 1.85 in both the non-academic and comprehensive schools. The distributions overlap strongly across all three tracks (Appendix [Figure E1](#)). Differences are more pronounced for achievement (Appendix [Figure E2](#)) and socioeconomic background. Academic-track students have higher average prior and later grades in both subjects and the highest parental SES, whereas parental SES is lowest in the non-academic track.¹⁷

Table 2. Track-specific sample composition, school-related stress, achievement, and socioeconomic background

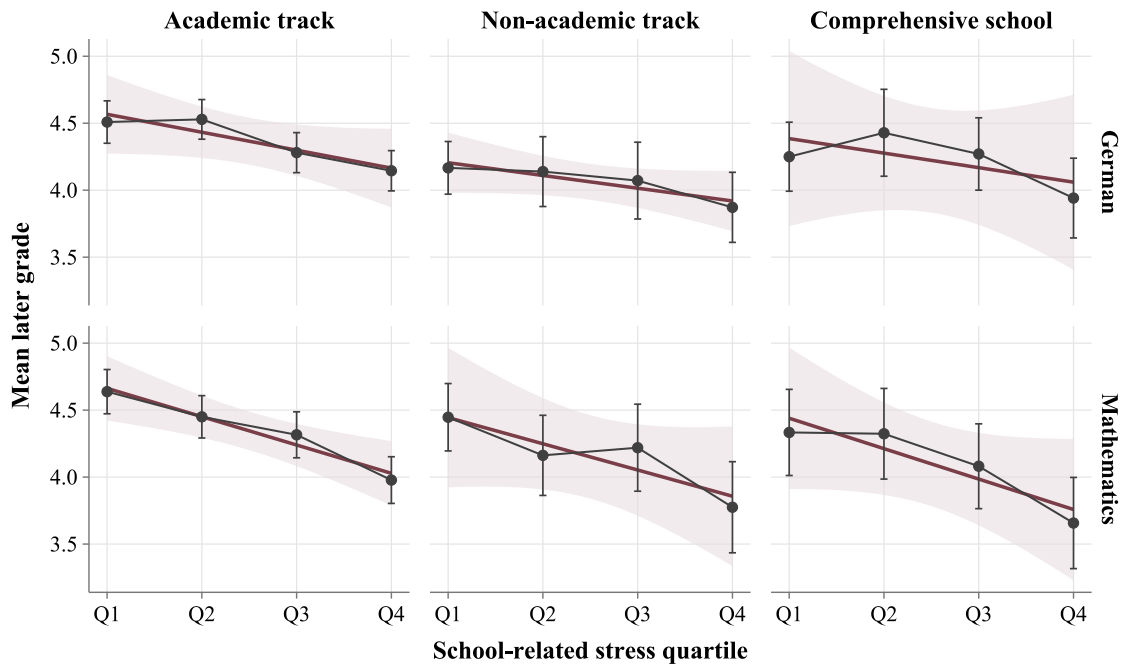
	School Type	N	N (Family)	School-related Stress		Prior Grades		Later Grades		SES	
				M	SD	M	SD	M	SD	M	SD
German	Academic track	526	324	1.92	0.52	4.69	0.8	4.36	0.9	66.07	16.44
	Non-academic track	165	111	1.85	0.52	4.13	0.75	4.07	0.82	53.96	18.12
	Comprehensive school	154	99	1.85	0.51	4.31	0.71	4.23	0.91	58.63	17.43
Math	Academic track	524	322	1.92	0.52	4.76	0.89	4.34	1.01	66	16.41
	Non-academic track	165	111	1.85	0.52	4.14	0.88	4.16	1.01	53.96	18.12
	Comprehensive school	154	98	1.85	0.51	4.37	0.93	4.12	1.07	58.9	17.38

Note. N denotes the number of students and N (Family) the number of unique families represented in each school-track group. Grades are reverse-coded. M = mean; SD = standard deviation.

[Figure 2](#) combines these descriptive patterns by showing later grades across quartiles of school-related stress within each school track. Later grades generally decrease across higher stress quartiles, with a more consistent pattern in mathematics. The overall pattern is also negative for German, although less regular among students in comprehensive schools. The descriptive results therefore indicate a negative stress-achievement gradient in both subjects, without an evident systematic difference in this gradient across school tracks.

¹⁷ These are descriptive group differences and do not indicate whether the association between school-related stress and achievement differs across tracks.

Figure 2. Later achievement across school-related stress quartiles by school track and subject



Note. Points show quartile-specific mean later grades; error bars indicate 95% confidence intervals around these means. The fitted line shows the linear descriptive trend across stress quartiles, with the shaded area indicating its 95% confidence interval. Q1 represents the lowest and Q4 the highest school-related stress quartile. Grades are reverse-coded. The figure is descriptive and unadjusted.

4.2. Main Longitudinal Analyses

Building on the descriptive results, the main longitudinal analyses examine the stress-achievement association (H1), its variation across school tracks (H2), its change after accounting for contextual conditions (RQ3), and prospective associations with later school-related stress (RQ4).

4.2.1. School-Related Stress and Later Achievement (H1)

[Tables 3](#) and [4](#) report the three-wave LDV models for German and mathematics. In the core model (M1), higher school-related stress at T1 is associated with lower later grades after prior subject-specific achievement, age, sex, and cohort are taken into account. The association is negative for both German ($b = -0.27$, $p < .001$) and mathematics ($b = -0.31$, $p < .001$). Adding school track in M2 leaves these estimates virtually unchanged (German: $b = -0.26$, $p < .001$; mathematics: $b = -0.31$, $p < .001$). H1 is therefore

supported for both subjects.¹⁸ The moderation and contextual extensions in M3 and M4 are examined in the following section.

Table 3. Main three-wave LDV models predicting later German grades

Variable	M1		M2		M3		M4	
	b	(SE)	b	(SE)	b	(SE)	b	(SE)
(Intercept)	2.36***	(0.26)	2.35***	(0.27)	2.37***	(0.27)	2.20***	(0.32)
School-related stress	-0.27***	(0.06)	-0.26***	(0.06)	-0.30***	(0.07)	-0.16*	(0.07)
Prior grade	0.42***	(0.04)	0.42***	(0.04)	0.42***	(0.04)	0.32***	(0.04)
Age	0.07**	(0.03)	0.07**	(0.03)	0.07**	(0.03)	0.05*	(0.02)
Sex (ref.: female)								
Male	-0.34***	(0.07)	-0.34***	(0.06)	-0.34***	(0.06)	-0.15*	(0.07)
Cohort (ref.: Cohort 1)								
Cohort 2	0.25	(0.20)	0.25	(0.19)	0.24	(0.20)	0.15	(0.16)
Cohort 3	-0.13	(0.24)	-0.14	(0.24)	-0.14	(0.24)	-0.12	(0.20)
Cohort 4	0.33	(0.29)	0.34	(0.29)	0.33	(0.29)	0.14	(0.24)
School track								
(ref.: Academic track)								
Non-academic track			-0.04	(0.08)	-0.03	(0.08)	0.02	(0.08)
Comprehensive school			0.09	(0.09)	0.09	(0.09)	0.10	(0.09)
Stress × school track								
(ref.: Academic track)								
Stress × non-academic track					0.12	(0.13)	0.14	(0.12)
Stress × comprehensive school					0.06	(0.16)	0.07	(0.15)
Contextual covariates								
Parental SES							0.00	(0.00)
Parental expectations							-0.04	(0.04)
Parental sanctions							-0.19***	(0.03)
Academic self-concept							0.19***	(0.06)
Intrinsic motivation							0.04	(0.05)
Learning motivation							0.10*	(0.04)
Number of friends							0.00	(0.00)
N	795		795		795		795	
R²	0.23		0.23		0.23		0.32	
Adjusted R²	0.22		0.22		0.22		0.30	

Note. Entries are unstandardized OLS coefficients (b) with family-clustered standard errors in parentheses. Grades are reverse-coded. Female, Cohort 1, and the academic track are the reference categories. School-related stress and age are mean-centered in all models; parental SES is mean-centered in M4. * $p < .05$, ** $p < .01$, *** $p < .001$.

¹⁸ Supplementary two-wave models ([Appendix F](#)) show the same overall negative association without adjustment for prior achievement. Supplementary cross-sectional analyses ([Appendix G](#)) likewise show a negative association, although stress and achievement are measured at the same wave in these models. These specifications are treated as supplementary checks, while the main interpretation is based on the three-wave LDV models.

Table 4. Main three-wave LDV models predicting later mathematics grades

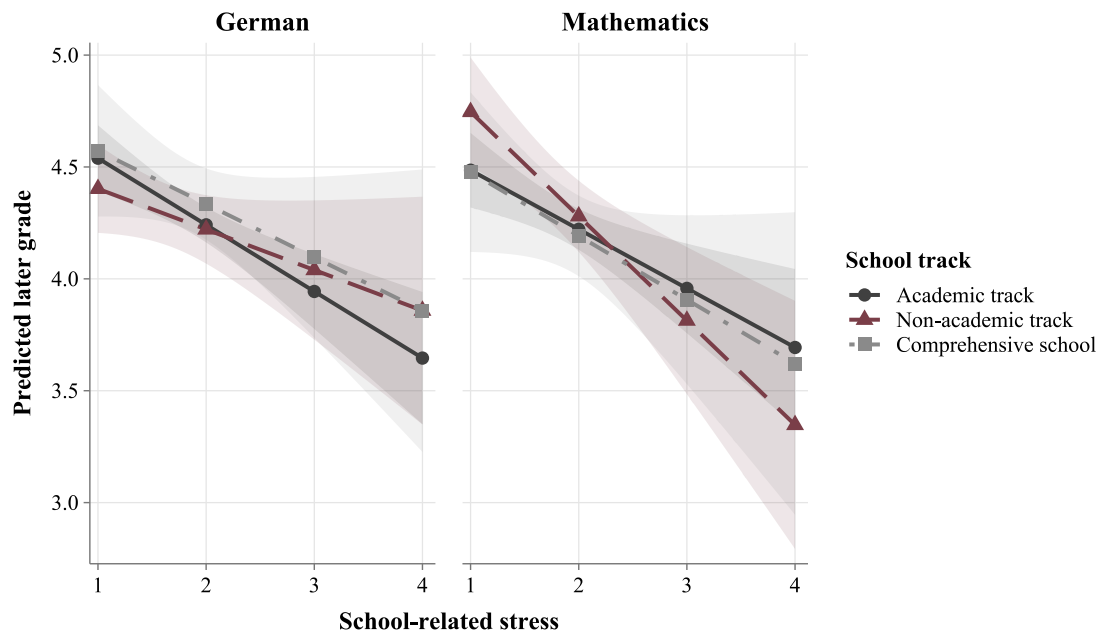
Variable	M1		M2		M3		M4	
	b	(SE)	b	(SE)	b	(SE)	b	(SE)
(Intercept)	1.89***	(0.27)	1.85***	(0.29)	1.82***	(0.29)	1.42***	(0.38)
School-related stress	-0.31***	(0.06)	-0.31***	(0.06)	-0.26**	(0.08)	-0.14°	(0.07)
Prior grade	0.47***	(0.04)	0.47***	(0.04)	0.47***	(0.04)	0.31***	(0.04)
Age	-0.03	(0.03)	-0.03	(0.03)	-0.03	(0.03)	-0.03	(0.03)
Sex (ref.: female)								
Male	-0.04	(0.07)	-0.04	(0.07)	-0.04	(0.07)	-0.05	(0.08)
Cohort (ref.: Cohort 1)								
Cohort 2	0.24	(0.22)	0.24	(0.22)	0.26	(0.22)	0.18	(0.21)
Cohort 3	0.51°	(0.27)	0.51°	(0.27)	0.52°	(0.27)	0.42°	(0.25)
Cohort 4	0.20	(0.42)	0.21	(0.42)	0.23	(0.42)	0.27	(0.47)
School track (ref.: Academic track)								
Non-academic track			0.09	(0.09)	0.08	(0.09)	0.06	(0.10)
Comprehensive school			-0.03	(0.10)	-0.03	(0.10)	-0.01	(0.10)
Stress × school track (ref.: Academic track)								
Stress × non-academic track					-0.20	(0.15)	-0.12	(0.15)
Stress × comprehensive school					-0.02	(0.18)	0.13	(0.16)
Contextual covariates								
Parental SES							0.00	(0.00)
Parental expectations							-0.02	(0.04)
Parental sanctions							-0.12***	(0.04)
Academic self-concept							0.34***	(0.06)
Intrinsic motivation							0.04	(0.05)
Learning motivation							0.07	(0.05)
Number of friends							-0.01	(0.00)
N	793		793		793		793	
R ²	0.23		0.23		0.23		0.34	
Adjusted R ²	0.22		0.22		0.22		0.32	

Note. Entries are unstandardized OLS coefficients (b) with family-clustered standard errors in parentheses. Grades are reverse-coded. Female, Cohort 1, and the academic track are the reference categories. School-related stress and age are mean-centered in all models; parental SES is mean-centered in M4. ° $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

4.2.2. School-Track Moderation (H2)

The moderation models provide no evidence that the association between school-related stress and later achievement differs systematically across school tracks. Consistent with the analytical strategy outlined in [Section 3.3](#), moderation is assessed primarily through joint tests of the two interaction terms. In M3, the joint Wald tests of the two interaction terms are non-significant for both German, $\chi^2(2) = 0.87$, $p = .648$, and mathematics, $\chi^2(2) = 1.90$, $p = .387$. The conclusion remains unchanged in M4 after the contextual covariates are added (German: $\chi^2(2) = 1.43$, $p = .490$; mathematics: $\chi^2(2) = 1.53$, $p = .464$; [Appendix Table D1](#)).

Figure 3. Adjusted predicted later grades by school-related stress and school track



Note. Predicted German and mathematics grades are based on M3 and control for prior subject-specific achievement, age, sex, and cohort. Shaded areas indicate 95% confidence intervals based on family-clustered standard errors. Grades are reverse-coded. The figure visualizes the fitted moderation model; formal tests of school-track differences are reported in Appendix [Table D1](#).

[Figure 3](#) visualizes the adjusted predictions from M3. Later grades decline with increasing school-related stress across all three school tracks in both subjects, while the estimated gradients differ only modestly relative to their uncertainty. Supplementary simple slopes and pairwise slope contrasts reported in [Appendix D](#) provide the same overall pattern. H2 is therefore not supported.¹⁹

4.2.3. Stress-Achievement Association Across Contexts (RQ3)

RQ3 examines how the stress-achievement association changes when school, family, social, and personal conditions are added to otherwise identical same-sample baseline models ([Tables 5](#) and [6](#)).²⁰ The extent of change differs clearly across the four contextual domains.

Accounting for the *school context* produces only modest changes. For German, the school-related stress coefficient decreases from $b = -0.26$ in the corresponding baseline model (M0-A) to $b = -0.20$ after adding school climate and teacher-student interaction

¹⁹ Supplementary analyses in [Appendices F](#) and [G](#) likewise provide no robust evidence that the school-related stress-achievement association differs systematically across school tracks.

²⁰ Attenuation of the stress coefficient indicates that part of the observed association overlaps with conditions in the respective context, whereas a largely stable coefficient suggests that the association remains comparatively independent of the measured indicators in that domain.

(M1-A). For mathematics, the coefficient decreases from $b = -0.30$ to $b = -0.27$. The stress coefficient remains statistically significant in both subjects. Teacher-student interaction is positively associated with later German grades ($b = 0.20$, $p < .01$), whereas neither school-context indicator is associated with later mathematics grades.

The *family context* produces stronger attenuation.²¹ For German, the stress coefficient decreases from $b = -0.23$ to $b = -0.10$ and is only marginally significant after the family conditions are added. For mathematics, it decreases from $b = -0.28$ to $b = -0.20$ but remains statistically significant. Parental sanctions are negatively associated with later achievement in both subjects (German: $b = -0.20$, $p < .001$; mathematics: $b = -0.17$, $p < .001$), while household chaos is additionally negatively associated with later German grades ($b = -0.10$, $p < .05$).

By contrast, the *social context* shows little change in the stress coefficient. For German, it changes from $b = -0.27$ to $b = -0.29$ after number of friends, loneliness, and bullying exposure are added; for mathematics, it changes from $b = -0.32$ to $b = -0.29$. The stress coefficient remains statistically significant in both subjects, and none of the social-context variables is significantly associated with later grades in the corresponding models.

The strongest attenuation occurs after accounting for the *personal context*. For German, the stress coefficient decreases from $b = -0.23$ to $b = -0.06$ and is no longer statistically significant. For mathematics, it decreases from $b = -0.26$ to $b = -0.13$ and remains only marginally significant. Academic self-concept is positively associated with later achievement in both subjects (German: $b = 0.24$, $p < .001$; mathematics: $b = 0.38$, $p < .001$), while self-regulation is additionally associated with German grades ($b = 0.23$, $p < .001$).

²¹ Parental SES is already included in all block-specific baseline models and is therefore not part of the incremental family-context adjustment reported here.

Table 5. Block-specific context models for German grades

Outcome: German grades																
Variable	School context models			Family context models			Social context models			Personal context models						
	798 (459)			716 (428)			735 (438)			588 (391)						
	M0-A Baseline	M1-A		M0-B Baseline	M1-B		M0-C Baseline	M1-C		M0-D Baseline	M1-D					
	b	(SE)	b	(SE)	b	(SE)	b	(SE)	b	(SE)	b	(SE)				
(Intercept)	2.48***	(0.26)	1.93***	(0.32)	2.39***	(0.29)	3.24***	(0.40)	2.33***	(0.28)	2.24***	(0.31)	2.38***	(0.32)	0.76*	(0.37)
School-related stress	-0.26***	(0.06)	-0.20***	(0.06)	-0.23***	(0.07)	-0.10°	(0.06)	-0.27***	(0.06)	-0.29***	(0.06)	-0.23***	(0.06)	-0.06	(0.06)
Prior grade	0.41***	(0.04)	0.41***	(0.04)	0.42***	(0.04)	0.38***	(0.04)	0.41***	(0.04)	0.41***	(0.04)	0.43***	(0.04)	0.35***	(0.04)
Age	0.08**	(0.03)	0.08**	(0.03)	0.08**	(0.03)	0.07**	(0.03)	0.07*	(0.03)	0.07*	(0.03)	0.07*	(0.03)	0.07***	(0.03)
Sex (ref.: female)																
Male	-0.36***	(0.07)	-0.34***	(0.07)	-0.37***	(0.07)	-0.26***	(0.07)	-0.35***	(0.07)	-0.34***	(0.07)	-0.35***	(0.07)	-0.25***	(0.07)
Cohort (ref.: Cohort 1)																
Cohort 2	0.16	(0.18)	0.16	(0.17)	0.21	(0.22)	0.06	(0.20)	0.32	(0.22)	0.34	(0.23)	0.19	(0.25)	0.13	(0.23)
Cohort 3	-0.22	(0.23)	-0.21	(0.22)	-0.32	(0.27)	-0.51*	(0.25)	-0.08	(0.28)	-0.04	(0.28)	-0.21	(0.30)	-0.14	(0.26)
Cohort 4	0.25	(0.29)	0.26	(0.31)	0.57	(0.41)	0.31	(0.35)	0.52	(0.35)	0.55	(0.35)	0.38	(0.36)	0.36	(0.28)
School track																
(ref.: Academic track)																
Non-academic track	-0.01	(0.09)	-0.02	(0.09)	0.02	(0.09)	0.06	(0.09)	0.01	(0.09)	-0.01	(0.09)	-0.03	(0.10)	-0.04	(0.09)
Comprehensive school	0.08	(0.09)	0.07	(0.09)	0.14	(0.09)	0.17°	(0.09)	0.13	(0.09)	0.13	(0.09)	0.12	(0.10)	0.09	(0.10)
Parental SES	0.00	(0.00)	0.00	(0.00)	0.00	(0.00)	0.00	(0.00)	0.00	(0.00)	0.00	(0.00)	0.00	(0.00)	0.00	(0.00)
School context																
School climate			-0.04	(0.07)												
Teacher-student interaction			0.20**	(0.08)												
Family context																
Parental expectations					0.03	(0.04)										
Parental autonomy support					-0.01	(0.05)										
Parental sanctions					-0.20***	(0.03)										
Household chaos					-0.10*	(0.04)										
Social context																
Number of friends											-0.01	(0.01)				
Loneliness											0.04	(0.04)				
Bullying exposure											0.03	(0.06)				
Personal context																
Academic self-concept															0.24***	(0.06)
Intrinsic motivation															0.03	(0.05)
Learning motivation															0.02	(0.06)
Self-efficacy															0.07	(0.07)
Self-regulation															0.23***	(0.05)
R²	0.23		0.24		0.24		0.23		0.29		0.23		0.24		0.33	
Adjusted R²	0.22		0.23		0.23		0.22		0.28		0.22		0.22		0.32	

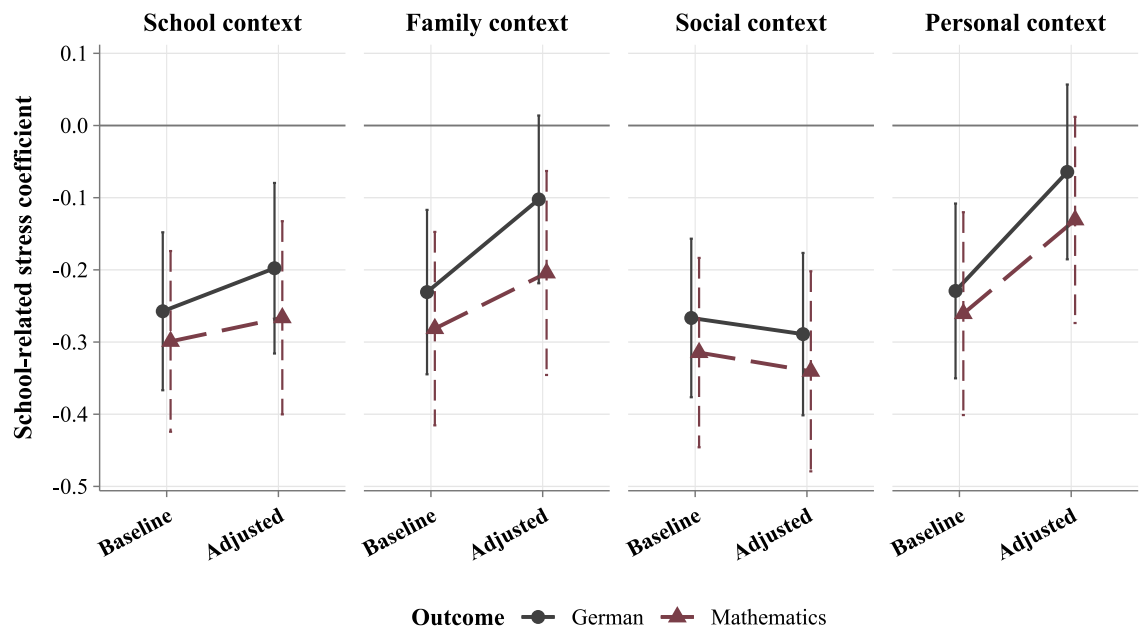
Note. Entries are unstandardized OLS coefficients (b) with family-clustered standard errors in parentheses. Each M0/M1 pair is estimated on the same complete-case sample; N (families) reports the number of students and unique family clusters for the respective context block. Grades are reverse-coded. Female, Cohort 1, and the academic track are the reference categories. School-related stress, age, and parental SES are mean-centered. ° p < .10, * p < .05, ** p < .01, *** p < .001.

Table 6. Block-specific context models for mathematics grades

Outcome: Mathematics grades												
Variable	School context models			Family context models			Social context models			Personal context models		
	796 (456)			714 (425)			734 (436)			587 (389)		
	M0-A Baseline	M1-A		M0-B Baseline	M1-B		M0-C Baseline	M1-C		M0-D Baseline	M1-D	
	b	(SE)	b	(SE)	b	(SE)	b	(SE)	b	(SE)	b	(SE)
(Intercept)	1.85***	(0.28)	1.53***	(0.36)	1.81***	(0.46)	1.64***	(0.33)	1.49***	(0.34)	1.70***	(0.35)
School-related stress	-0.30***	(0.06)	-0.27***	(0.07)	-0.20***	(0.07)	-0.32***	(0.07)	-0.29***	(0.09)	-0.26***	(0.07)
Prior grade	0.48***	(0.04)	0.48***	(0.04)	0.46***	(0.04)	0.48***	(0.04)	0.48***	(0.04)	0.50***	(0.04)
Age	-0.01	(0.03)	-0.01	(0.03)	-0.01	(0.04)	0.00	(0.03)	0.00	(0.03)	0.01	(0.04)
Sex (ref.: female)												
Male	-0.08	(0.07)	-0.07	(0.08)	0.02	(0.08)	-0.06	(0.08)	-0.05	(0.08)	-0.08	(0.09)
Cohort (ref.: Cohort 1)												
Cohort 2	0.20	(0.22)	0.18	(0.22)	0.20	(0.24)	0.43°	(0.26)	0.46°	(0.25)	0.28	(0.28)
Cohort 3	0.41	(0.27)	0.39	(0.27)	0.30	(0.30)	0.54°	(0.30)	0.58°	(0.30)	0.38	(0.32)
Cohort 4	0.16	(0.43)	0.16	(0.47)	-0.12	(0.49)	0.55	(0.54)	0.59	(0.50)	0.43	(0.56)
School track (ref.: Academic track)												
Non-academic track	0.14	(0.10)	0.13	(0.10)	0.16	(0.10)	0.16	(0.10)	0.14	(0.10)	0.17	(0.10)
Comprehensive school	-0.03	(0.10)	-0.03	(0.11)	0.00	(0.11)	0.02	(0.11)	0.02	(0.11)	-0.05	(0.12)
Parental SES	0.00	(0.00)	0.00	(0.00)	0.00	(0.00)	0.00	(0.00)	0.00	(0.00)	0.00°	(0.00)
School context												
School climate			0.06	(0.07)								
Teacher-student interaction			0.05	(0.09)								
Family context												
Parental expectations					0.07	(0.05)						
Parental autonomy support					-0.11°	(0.06)						
Parental sanctions					-0.17***	(0.04)						
Household chaos					-0.05	(0.05)						
Social context												
Number of friends									-0.01	(0.01)		
Loneliness									0.04	(0.04)		
Bullying exposure									0.04	(0.08)		
Personal context												
Academic self-concept											0.38***	(0.06)
Intrinsic motivation											0.01	(0.05)
Learning motivation											0.10°	(0.06)
Self-efficacy											-0.07	(0.07)
Self-regulation											0.06	(0.05)
R²	0.24		0.24		0.23		0.23		0.23		0.25	
Adjusted R²	0.23		0.23		0.22		0.22		0.22		0.23	

Note. Entries are unstandardized OLS coefficients (b) with family-clustered standard errors in parentheses. Each M0/M1 pair is estimated on the same complete-case sample; N (families) reports the number of students and unique family clusters for the respective context block. Grades are reverse-coded. Female, Cohort 1, and the academic track are the reference categories. School-related stress, age, and parental SES are mean-centered. ° p < .10, * p < .05, ** p < .01, *** p < .001.

Figure 4. School-related stress coefficients before and after block-specific contextual adjustment



Note. Points show estimated school-related stress coefficients from the block-specific same-sample baseline and context-adjusted models for German and mathematics. Intervals indicate 95% confidence intervals based on family-clustered standard errors. Each baseline model is estimated on the same analytical sample as its corresponding context-adjusted model. Negative coefficients indicate lower later achievement at higher levels of school-related stress.

[Figure 4](#) summarizes the block-specific changes in the school-related stress coefficient. Overall, the association changes most strongly after personal-context adjustment, followed by the family context, whereas accounting for the measured school and social conditions leaves it comparatively stable. The attenuation is generally more pronounced for German than for mathematics.

4.2.4. Later Stress Across Contexts and Tracks (RQ4)

RQ4 examines whether contextual conditions at T1 are associated with school-related stress at T2 after accounting for students' earlier school-related stress, and whether these associations differ across school tracks ([Tables 7](#) and [8](#)). Across the four interaction models, prior school-related stress remains strongly associated with later stress ($b = 0.46 - 0.47$), indicating substantial temporal stability over the observation period. Because both stress measures are standardized, students who report stress one standard deviation above others at T1 also tend to report roughly half a standard deviation higher stress at T2, conditional on the remaining model covariates.

Beyond this temporal stability, only a small number of contextual coefficients reach conventional significance levels. In the academic-track reference group of the school

context model, teacher-student interaction shows a marginal negative association with later stress ($b = -0.07$, $p < .10$; [Table 7](#)), whereas school climate shows no corresponding association. In the social-context model, the coefficient for loneliness is negative ($b = -0.07$, $p < .05$), while bullying exposure is positively associated with later stress ($b = 0.10$, $p < .05$). Neither the family- nor the personal-context model shows clear prospective associations within the academic-track reference group ([Table 8](#)).

Evidence that these contextual associations differ systematically across school tracks is limited. The joint Wald tests are non-significant for the school context, $\chi^2(4) = 4.41$, $p = .661$, the family context, $\chi^2(10) = 11.42$, $p = .326$, and the personal context, $\chi^2(10) = 11.35$, $p = .331$. The social-context block provides the strongest indication of track heterogeneity, but the joint test remains marginal, $\chi^2(6) = 12.29$, $p = .056$ (Appendix [Table H2](#)). Given the number of contextual coefficients and interaction terms examined in RQ4, isolated coefficient-level findings should therefore be interpreted cautiously.

[Figure 5](#) displays the school-track-specific slopes for contextual variables for which at least one track-specific slope has $p < .10$. All three track-specific estimates are shown for each selected variable so that these isolated associations can be compared across tracks. The clearest differences appear in the social context, alongside individual patterns involving teacher-student interaction, parental SES, household chaos, and learning motivation. However, the complete set of estimates in Appendix [Figure H1](#) shows substantial uncertainty around most slopes. [Figure 5](#) should therefore be interpreted as an exploratory illustration rather than as evidence of a systematic track-specific pattern.

Table 7. Prospective models predicting later school-related stress: school and social contexts

Variable	School context models 1129 (579)				Social context models 1039 (560)			
	Baseline		Context × school track		Baseline		Context × school track	
	b	(SE)	b	(SE)	b	(SE)	b	(SE)
(Intercept)	0.29°	(0.17)	0.27	(0.17)	0.40*	(0.16)	0.40*	(0.17)
Prior school-related stress (T1)	0.48***	(0.03)	0.46***	(0.03)	0.49***	(0.03)	0.47***	(0.03)
Age	0.10***	(0.03)	0.09**	(0.03)	0.10***	(0.03)	0.11***	(0.03)
Sex (ref.: female)								
Male	-0.41***	(0.06)	-0.43***	(0.06)	-0.41***	(0.06)	-0.41***	(0.06)
Cohort (ref.: Cohort 1)								
Cohort 2	-0.08	(0.17)	-0.05	(0.17)	-0.21	(0.16)	-0.20	(0.17)
Cohort 3	-0.09	(0.21)	-0.04	(0.21)	-0.24	(0.21)	-0.21	(0.22)
Cohort 4	0.45	(0.41)	0.35	(0.41)	0.33	(0.41)	0.38	(0.30)
School track (ref.: Academic track)								
Non-academic track	-0.06	(0.07)	-0.06	(0.07)	-0.05	(0.07)	-0.06	(0.07)
Comprehensive school	0.06	(0.08)	0.06	(0.08)	0.06	(0.08)	0.05	(0.08)
School context								
School climate			-0.02	(0.04)				
Teacher-student interaction			-0.07°	(0.04)				
School context × school track (ref.: Academic track)								
Non-academic track × school climate			-0.09	(0.09)				
Comprehensive school × school climate			0.06	(0.09)				
Non-academic track × teacher-student interaction			0.02	(0.10)				
Comprehensive school × teacher-student interaction			0.02	(0.10)				
Social context								
Number of friends							-0.04	(0.04)
Loneliness							-0.07*	(0.03)
Bullying exposure							0.10*	(0.05)
Social context × school track (ref.: Academic track)								
Non-academic track × number of friends							-0.01	(0.07)
Comprehensive school × number of friends							0.15*	(0.06)
Non-academic track × loneliness							0.14°	(0.08)
Comprehensive school × loneliness							0.07	(0.07)
Non-academic track × bullying exposure							-0.08	(0.08)
Comprehensive school × bullying exposure							-0.09	(0.08)
R²	0.31		0.32		0.32		0.33	
Adjusted R²	0.30		0.31		0.31		0.32	

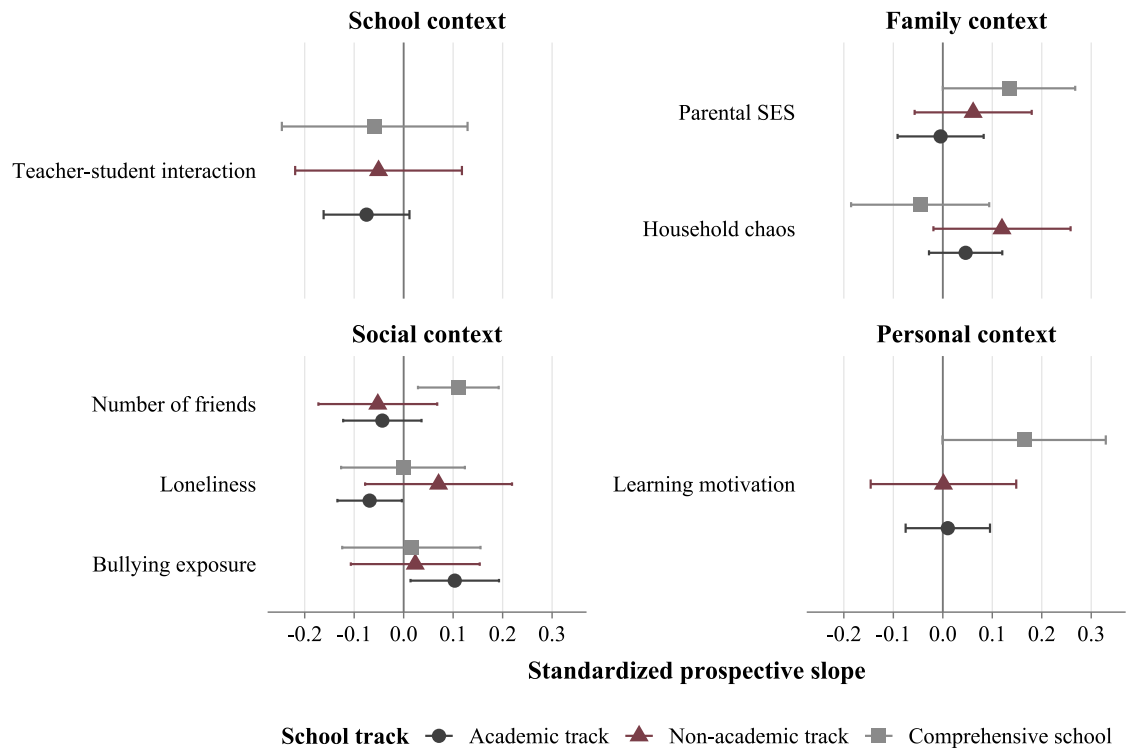
Note. Entries are OLS coefficients (b) with family-clustered standard errors in parentheses. Within each context block, the baseline and context × school-track models are estimated on the same complete-case sample; N (families) reports the number of students and unique family clusters. School-related stress at T1 and T2, age, and all continuous contextual predictors are standardized. Female, Cohort 1, and the academic track are the reference categories. In the interaction models, contextual main effects refer to the academic-track reference group. ° $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 8. Prospective models predicting later school-related stress: family and personal contexts

Variable	Family context models 973 (521)				Personal context models 798 (500)			
	Baseline		Context × school track		Baseline		Context × school track	
	b	(SE)	b	(SE)	b	(SE)	b	(SE)
(Intercept)	0.24	(0.19)	0.20	(0.20)	0.48**	(0.17)	0.47*	(0.18)
Prior school-related stress (T1)	0.47***	(0.03)	0.47***	(0.04)	0.49***	(0.04)	0.46***	(0.04)
Age	0.08**	(0.03)	0.09**	(0.03)	0.09**	(0.03)	0.10**	(0.03)
Sex (ref.: female)								
Male	-0.38***	(0.06)	-0.38***	(0.06)	-0.36***	(0.06)	-0.36***	(0.07)
Cohort (ref.: Cohort 1)								
Cohort 2	-0.03	(0.19)	0.00	(0.20)	-0.32°	(0.17)	-0.29	(0.18)
Cohort 3	-0.18	(0.23)	-0.16	(0.23)	-0.28	(0.23)	-0.29	(0.23)
Cohort 4	1.01***	(0.19)	1.16***	(0.22)	0.25	(0.41)	0.17	(0.42)
School track (ref.: Academic track)								
Non-academic track	-0.07	(0.07)	-0.05	(0.08)	-0.11	(0.08)	-0.14°	(0.09)
Comprehensive school	0.03	(0.08)	0.06	(0.08)	0.09	(0.09)	0.07	(0.09)
Family context								
Parental SES			0.00	(0.04)				
Parental expectations			0.01	(0.04)				
Parental autonomy support			0.06	(0.05)				
Parental sanctions			-0.01	(0.04)				
Household chaos			0.05	(0.04)				
Family context × school track (ref.: Academic track)								
Non-academic track × parental SES			0.07	(0.07)				
Comprehensive school × parental SES			0.14°	(0.08)				
Non-academic track × parental expectations			0.14	(0.11)				
Comprehensive school × parental expectations			0.05	(0.09)				
Non-academic track × parental autonomy support			-0.07	(0.08)				
Comprehensive school × parental autonomy support			-0.11	(0.07)				
Non-academic track × parental sanctions			-0.04	(0.07)				
Comprehensive school × parental sanctions			0.09	(0.08)				
Non-academic track × household chaos			0.07	(0.08)				
Comprehensive school × household chaos			-0.09	(0.08)				
Personal context								
Academic self-concept							-0.07	(0.05)
Intrinsic motivation							-0.08	(0.05)
Learning motivation							0.01	(0.04)
Self-efficacy							0.00	(0.05)
Self-regulation							0.04	(0.04)
Personal context × school track (ref.: Academic track)								
Non-academic track × academic self-concept							0.07	(0.10)
Comprehensive school × academic self-concept							0.04	(0.09)
Non-academic track × intrinsic motivation							0.09	(0.11)
Comprehensive school × intrinsic motivation							0.04	(0.11)
Non-academic track × learning motivation							-0.01	(0.09)
Comprehensive school × learning motivation							0.15	(0.09)
Non-academic track × self-efficacy							-0.12	(0.09)
Comprehensive school × self-efficacy							-0.10	(0.12)
Non-academic track × self-regulation							0.01	(0.09)
Comprehensive school × self-regulation							-0.13	(0.10)
R²	0.30		0.31		0.32		0.34	
Adjusted R²	0.29		0.30		0.31		0.32	

Note. Entries are OLS coefficients (b) with family-clustered standard errors in parentheses. Within each context block, the baseline and context × school-track models are estimated on the same complete-case sample; N (families) reports the number of students and unique family clusters. School-related stress at T1 and T2, age, and all continuous contextual predictors are standardized. Female, Cohort 1, and the academic track are the reference categories. In the interaction models, contextual main effects refer to the academic-track reference group. ° $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

Figure 5. Selected school-track-specific prospective contextual associations with later school-related stress



Note. Points show standardized school-track-specific prospective slopes with 95% confidence intervals based on family-clustered standard errors. Contextual variables are included when at least one track-specific slope has $p < .10$. Models predict standardized school-related stress at T2 while controlling for standardized school-related stress at T1, age, sex, cohort, and school track. Continuous contextual predictors are standardized.

Taken together, the analyses support H1, showing a negative association between school-related stress and later achievement, whereas H2, which expected this association to differ across school tracks, is not supported. RQ3 indicates that the stress-achievement association is contextually embedded, particularly in personal and family conditions. RQ4, which shifts the focus to later school-related stress, points to substantial temporal stability and only limited exploratory evidence that contextual associations differ systematically across school tracks.

5. Discussion and Conclusion

This chapter interprets the empirical findings in relation to the theoretical framework and previous research. It first discusses the findings for H1, H2, RQ3, and RQ4, then addresses the study's limitations and directions for future research, and concludes by summarizing its main findings and contribution.

5.1. Interpretation of Findings

School-related stress and later achievement (H1)

The central finding for H1 is that students reporting higher school-related stress subsequently received poorer grades in both German and mathematics, even after their earlier achievement in the respective subject was taken into account. The association therefore cannot be reduced to achievement differences that were already present before stress was measured. This is consistent with previous longitudinal findings from German secondary education ([Grützmacher & Raufelder, 2019](#); [Becker & Börnert-Ringleb, 2026](#)). The present study adds to this evidence by examining subject-specific report-card grades and controlling for earlier achievement in the same subject.

The finding is also consistent with the appraisal-based framework developed in [Section 2.3](#). Perceived school-related stress captures students' experience that school demands are difficult to manage relative to their available resources. Control-value theory provides one possible account of how such experiences may be connected to later achievement. Achievement-related emotions can interfere with attention, motivation, learning strategies, and self-regulated learning ([Pekrun, 2006](#)). These processes were not measured directly, however, so the present analyses cannot determine which mechanisms connect perceived school-related stress with later grades.

H1 is therefore supported for both German and mathematics. At the same time, the comparatively modest associations indicate that perceived school-related stress is only one of several factors related to later achievement. The findings thus support a prospective association with achievement without implying that stress alone determines subsequent performance. From the theoretical perspective outlined above, such modest associations are also plausible if the relationship between perceived stress and achievement operates partly through intermediate emotional, cognitive, or motivational processes that were not directly examined in this study.

School-track differences of the stress-achievement association (H2)

H2 is not supported. Although German secondary school tracks differ in student composition, instructional conditions, and learning opportunities, the association between perceived school-related stress and later achievement is broadly similar across academic, non-academic, and comprehensive school settings. This provides no evidence for the moderation implied by the stage-environment fit perspective, according to which differences in educational environments may alter the fit between adolescents' needs and the

demands and resources they encounter ([Eccles et al., 1993](#)).

This does not imply that school tracks provide equivalent developmental environments. Track-level differences in demands and resources may still shape students' exposure to stress or their appraisal of school demands without necessarily changing the association between perceived stress and subsequent achievement. Moreover, school-track membership captures relatively broad educational environments and may obscure variation in more proximal conditions within tracks, such as teaching quality, learning opportunities, or classroom competition ([Schiepe-Tiska, 2019](#); [Becker & Börnert-Ringleb, 2025](#)).

Contextual embeddedness of the stress-achievement association (RQ3)

Whereas H1 and H2 tested theoretically derived expectations, RQ3 and RQ4 extend the analysis to aspects of the framework for which theory and previous evidence did not provide sufficiently clear directional expectations. RQ3 first examines how the stress-achievement association changes when family, school, social, and personal conditions are taken into account.

The strongest attenuation occurs after accounting for the personal context. Academic self-concept, motivation, self-efficacy, and self-regulation are closely related to how students engage with schoolwork and perform academically; academic self-concept, for example, is reciprocally related to achievement over time ([Marsh & Martin, 2011](#)). This close connection to both perceived stress and achievement is consistent with the substantial reduction of the stress coefficient, particularly in German. Because these conditions were measured alongside stress at T1, however, the analyses cannot determine whether they precede stress, develop alongside it, or form part of the processes linking stress to later achievement.

The family block shows a similar, although less pronounced, attenuation. Because parental SES is already included in the corresponding baseline models, the additional reduction in the stress coefficient is associated with the family conditions added subsequently. Parental sanctions are the clearest family correlate of later achievement in both subjects, suggesting that students' experiences surrounding school performance at home overlap with part of the observed stress-achievement association. Longitudinal evidence likewise links parental support and negative parent-adolescent interactions to adolescents' responses to academic stressors ([Zimmer-Gembeck et al., 2023](#)).

By contrast, adding the measured school and social conditions changes the stress coefficient comparatively little. This does not imply that these contexts are unimportant

for academic achievement. Rather, within the present models, they account for little of the observed association between perceived school-related stress and later grades and may influence how stressful school demands are perceived or how students cope with them ([Hoferichter et al., 2022](#)).

The attenuation is generally stronger for German than for mathematics, especially in the family and personal models, but the available measures provide no clear explanation for this subject difference. More generally, the size of the coefficient changes should not be interpreted as a ranking of the four contextual domains. The blocks differ in both their composition and their proximity to academic achievement. RQ3 therefore shows where the observed stress-achievement association overlaps most strongly with the measured conditions, not which context is causally most important.

Stress Stability and Contextual Associations (RQ4)

RQ4 shifts the focus from achievement to the development of school-related stress itself. The most prominent finding is the substantial temporal continuity of perceived stress. Students reporting higher school-related stress at T1 also tend to report higher stress at T2 after the contextual conditions are taken into account. This closely corresponds with Becker and Börnert-Ringleb ([2026](#)), who likewise report strong autoregressive associations of school-related stress across three measurement waves.²² The present finding therefore suggests that differences in perceived school-related stress are not merely short-lived, but show considerable persistence across measurement occasions.

The contextual models provide only limited evidence on conditions associated with later stress. Most contextual conditions were measured at T1, at the same time as the initial stress measure. Because T1 stress is controlled when predicting T2 stress, relationships that are already reflected in students' initial stress levels may no longer appear as additional prospective associations. The generally small coefficients therefore do not imply that family, school, social, or personal conditions are unimportant for school-related stress. They show only that these conditions add little information about later stress once earlier stress is taken into account.

A similar picture emerges for school-track differences. The joint tests provide no clear evidence that the contextual associations with later stress systematically differ across academic, non-academic, and comprehensive school settings, although the social

²² At the same time, this should not be equated with chronic stress. Chronic stress refers to sustained or recurrent stressful conditions over an extended period, whereas two measurements of perceived school-related stress cannot establish such a temporal pattern on their own ([Epel et al., 2018](#)).

context shows a tentative indication of heterogeneity. Individual track-specific differences may therefore exist, but the present results do not support a broader pattern in which school track consistently alters how the measured contextual conditions are associated with later stress.

5.2. Limitations

Several limitations constrain the interpretation of the findings. First, the three-wave LDV design improves temporal ordering by measuring prior achievement at T0, school-related stress at T1, and later achievement at T2, but it does *not* identify causal effects. Unobserved individual, family, or school factors may affect both stress and achievement, and family-clustered standard errors do not remove such confounding. Moreover, the main achievement models examine the association from stress to later achievement and do not capture potentially reciprocal dynamics between both constructs ([Becker & Börnert-Ringleb, 2026](#)).

Second, the stress measure captures students' subjective school-related burden rather than the specific demands underlying it. H2 therefore examines whether the stress-achievement association differs across school tracks, not whether students in different tracks encounter different stressors. The seven-item index also combines perceived pressure with related experiences such as rumination and exhaustion. Although its reliability is acceptable and similar across tracks, the better fit of a two-factor solution suggests that it should be interpreted as a broad measure of perceived school-related stress rather than as a pure appraisal measure ([Epel et al., 2018](#); [Crosswell & Lockwood, 2020](#)). The absence of measurement-invariance tests across tracks and waves further limits stronger comparisons, and the study does not directly observe the emotional, cognitive, or motivational processes that may connect stress with achievement.

Third, the timing of the contextual measures limits what can be concluded from RQ3 and RQ4. In RQ3, most contextual conditions are measured at the same time as school-related stress. The reduction of the stress coefficient therefore shows overlap between these variables, but not mediation or a clear temporal sequence. In RQ4, the contextual conditions are again measured at T1, while earlier stress is controlled when predicting later stress. As a result, relationships that are already reflected in students' T1 stress may no longer appear as additional associations with T2 stress. Comparisons across the four contextual domains should also remain cautious because the blocks differ in their indicators and sample sizes. Finally, the large number of coefficients and interaction terms

means that individual RQ4 findings should be interpreted cautiously.

Fourth, the three school-track categories provide only a broad picture of German secondary education. They cannot capture differences between Länder, individual schools and classrooms, or variation within the same track. Students also do not enter school tracks at random; track placement is related to characteristics such as prior achievement and social background ([Baumert et al., 2006](#)). In addition, possible changes between school tracks from T1 to T2 are not considered here. The academic-track group is much larger than the other two groups, which makes smaller differences between tracks harder to detect. For these reasons, the non-significant interactions should not be interpreted as evidence that the stress-achievement association is identical across tracks. They only show that this study did not find clear and systematic differences between them.

Fifth, report-card grades are consequential educational outcomes and contain information relevant to students' subsequent educational performance ([Thorsen & Cliffordson, 2012](#)). However, they are assigned by teachers and may reflect grading standards that vary across schools and school tracks. The same numerical grade may therefore not correspond to the same level of competence across schools or school tracks ([Bowers, 2011](#)). The ordinal-logistic robustness analyses address whether treating grades as approximately metric changes the substantive conclusions, but they cannot resolve differences in grading standards. Standardized competence measures would therefore provide an important complementary outcome ([Hübner et al., 2020](#)).

More generally, the measures available in TwinLife were not specifically tailored to research on school-related stress and its educational contexts. The analyses therefore rely on the available measures and observation waves, which restrict the temporal alignment of stress and contextual conditions and the level of detail with which relevant educational contexts can be examined. Generalizability is additionally limited by the analytical sample. TwinLife is based on a probability sample of twin families and is not a population-representative sample of all German secondary-school students. The present samples are further restricted by longitudinal participation, available school-track information, and complete data for the respective models. Survey weights are not applied, and selective missingness cannot be excluded ([Lang & Kottwitz, 2020](#); [Krell et al., 2025](#)). The results should therefore be interpreted as associations within the observed longitudinal samples rather than as population-representative estimates.

5.3. Implications and Conclusion

This study examined how perceived school-related stress is associated with later academic achievement in German secondary education and whether this relationship differs across school tracks and broader contextual conditions. Higher school-related stress was associated with poorer subsequent German and mathematics grades even after earlier achievement was taken into account. At the same time, the association was modest and showed no systematic differences across academic, non-academic, and comprehensive school settings.

The stress-achievement association was reduced most strongly after personal and family conditions were considered, whereas the measured school and social conditions changed it less. School-related stress also showed substantial continuity over time. Once earlier stress was taken into account, only limited additional associations with the measured contextual conditions remained, and these did not differ systematically across school tracks.

Taken together, the findings suggest that school-related stress should not be understood as a problem confined to a particular school track. Its association with later achievement appears across different secondary-school settings, while students' personal and family conditions are closely connected to this relationship. Broad school-track categories alone therefore provide only limited information about the conditions under which school-related stress becomes relevant for academic development.

Future research should distinguish more clearly between different components of the stress process. The present data provide only a broad measure of perceived school-related stress and do not separately capture the demands students encounter, how these demands are appraised, and the emotional, cognitive, or behavioural responses that follow. Future studies should therefore combine measures of concrete school-related stressors with students' subjective appraisal of these demands and subsequent stress responses ([Epel et al., 2018](#); [Crosswell & Lockwood, 2020](#)). Such a design would allow researchers to examine more directly why similar school demands are experienced differently across students and which parts of the stress process are most closely related to later achievement. Repeated measurements of these components could additionally clarify how stressors, perceived stress, and academic outcomes develop in relation to one another over time.

Research on educational contexts should also move beyond broad school-track categories. More specific information on schools, classrooms, and instructional conditions could help identify which aspects of students' learning environments are related to stress

and its academic consequences (e.g., [Holzberger & Schiepe-Tiska, 2021](#)). Länder-specific institutional differences should likewise be considered rather than treating German secondary education as a uniform setting (e.g., [Matthewes, 2021](#)). Finally, combining report-card grades with standardized competence measures would help distinguish changes in academic competence from differences in grading practices across educational settings (e.g., [Hübner et al., 2020](#); [Westphal et al., 2021](#)). Larger and more balanced samples across school tracks would further improve the ability to detect smaller differences between these settings.

Because TwinLife includes monozygotic and dizygotic twin pairs, future research could use multivariate twin models to examine whether the association between school-related stress and academic achievement reflects overlapping genetic, shared environmental, and non-shared environmental influences, extending a similar approach used for mathematics anxiety and achievement ([Malanchini et al., 2020](#)).

The broader implication is that school-related stress should be studied as part of a process rather than as an isolated student characteristic. Distinguishing the demands students encounter, how they experience them, and the resources available to deal with them offers a more precise basis for understanding when and why school-related stress becomes consequential for educational development.

The findings also underline the need for more specific theoretical accounts of how perceived (school-related) stress is embedded in students' resources and educational environments. In particular, clearer expectations are needed about when and under which conditions these contexts alter the relationship between stress and academic outcomes.

Understanding school-related stress as part of students' broader educational development requires closer attention to the demands students encounter, how they experience these demands, and the resources available to deal with them. The central task for future research is therefore to identify more precisely the conditions and processes underlying the association between perceived school-related stress and later academic achievement.

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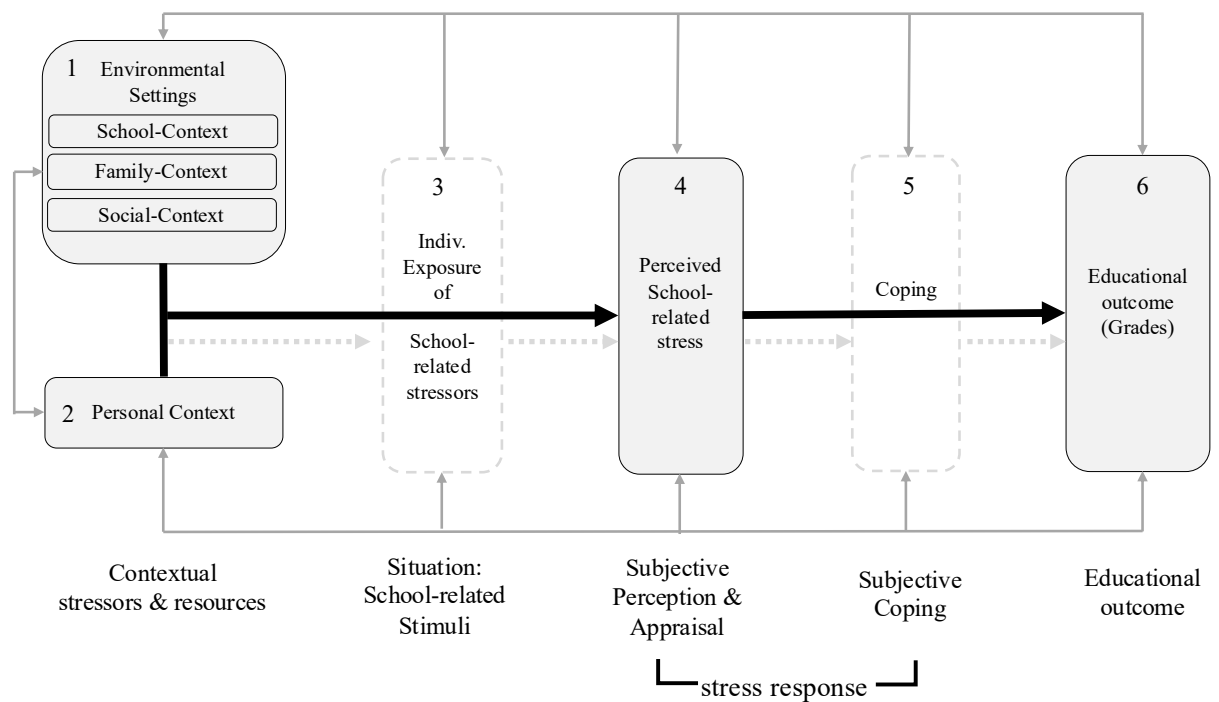
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Appendix A. Theoretical Background

Appendix A presents the broader conceptual model underlying [Figure 1](#). It situates perceived school-related stress within a wider process linking contextual conditions, appraisal, coping, and educational outcomes. The model serves as a theoretical organizing framework rather than as a fully tested causal model.

Figure A1. Extended theoretical framework of school-related stress, contextual stressors and resources, coping, and educational outcomes.



Note. Adapted from Moos and Holahan (2003), Morazes (2016), and Lazarus and Folkman (1984). The figure depicts theoretical pathways of the broader stress process. Stressor exposure, coping, and reciprocal pathways are not directly tested in the present study and are therefore not all retained in the empirically focused framework in [Figure 1](#).

Appendix B. Measurement, Item Composition, and Scale Diagnostics

Appendix B documents the construction and diagnostic assessment of the multi-item measures used in the analyses, together with the recoding of school track. The tables report item composition and scoring, scale reliability, the dimensionality check for perceived school-related stress, and the mapping of the original TwinLife school-type categories to the three analytical school-track groups.

Table B1. Item composition of the multi-item indices

	Construct	Item	TwinLife item label	Waves	Scale	Scoring	Rev.	
School	School-related stress	edu0901	Stress: tension	2/3/4/5	1=CD; 4=CA	Mean of all 7 items; higher = higher stress		
		edu0902	Stress: difficulties falling asleep					
		edu0903	Stress: I react very irritably					
		edu0904	Stress: think about school problems in my free time					
		edu0905	Stress: I'm exhausted					
		edu0906	Stress: the pressure at school is too high					
		edu0907	Stress: no time for other things					
	School climate	edu0700	School climate: good student-teacher relation	2/3/5		Mean of listed items; higher = more positive climate		
		edu0701	School climate: students' well-being is important					
		Teacher-student interaction	edu0800					Teacher interaction: fair treatment
	edu0801	Teacher interaction: extra help						
	edu0802	Teacher interaction: teachers listening students						
Family	Parental expectations	inv0100	Parental structure: aware of par. expectations of efforts in school	1/2/4		Mean of listed items; higher = stronger expectations		
		inv0101	Parental structure: aware of par. expectations for school performance					
	Parental autonomy support	inv0106	Parental autonomy: parents encourage to set and achieve goals	1/2/4	1=NC; 5=FC	Mean of listed items; higher = stronger autonomy support		
		inv0107	Parental autonomy: parents encourage to ask them questions					
		inv0108	Parental autonomy: parents encourage to ask questions in school					
	Parental sanctions	inv0109	Parental control: parents scold children over poor grades	1/2/4		Mean of listed items; higher = stronger sanctions/control		
		inv0110	Parental control: parents threaten punishment for poor grades					
		inv0111	Parental control: parents reproach children for poor grades					
	Personal	Household chaos	hoe0300	at home everything is chaotic.	1/2/5		Mean after reversing hoe0600; higher = more household chaos	Yes
		hoe0600	the atmosphere at home is quiet.					
Academic self-concept, general school		spa0200	Self-concept: talent in school in general	1/2/3			Yes	
		spa0201	Self-concept: ability for school in general					
		spa0202	Self-concept: problem-solving in school in general					
Academic self-concept, mathematics		spa0300	Self-concept: talent for mathematics	1/2/5		Mean after reversing negative item; higher = stronger self-concept	Yes	
		spa0301	Self-concept: ability in mathematics					
		spa0302	Self-concept: problem-solving in mathematics					
Academic self-concept, German		spa0400	Self-concept: talent for subject German	1/2/5			Yes	
		spa0401	Self-concept: ability in subject German					
	spa0402	Self-concept: problem-solving in German						
Social	Intrinsic motivation, school	imo0200	Intrinsic motivation: I like doing the things I learn at school	1/2/3				
		imo0201	Intrinsic motivation: School is fun					
		imo0202	Intrinsic motivation: Things that I learn (at school) are interesting.					
	Intrinsic motivation, mathematics	imo0300	Intrinsic motivation: I like doing math	1/2/5	1–5	Mean of listed items; higher = stronger intrinsic motivation		
		imo0301	Intrinsic motivation: Math is fun					
		imo0302	Intrinsic motivation: Math is interesting					
	Intrinsic motivation, German	imo0400	Intrinsic motivation: I like learning German	1/2/5				
		imo0401	Intrinsic motivation: German is fun					
		imo0402	Intrinsic motivation: German is interesting					
	Learning motivation	imo0500	Learning motivation: to learn interesting things	1/2/3/5		Mean of listed items; higher = stronger learning motivation		
imo0501		Learning motivation: understanding difficult things						
imo0502		Learning motivation: learning as much as possible.						
Self-efficacy	sef0100	relying on own abilities in difficult situations	1/2/3/5		Mean of listed items; higher = stronger self-efficacy			
	sef0101	ability to solve problems						
	sef0102	ability of solving tasks						
Self-regulation		srg0100	Self-regulation: new projects distract me from previous ones	2/4/5		Mean of original low self-regulation items, then reversed at index level; higher = stronger self-regulation	Index	
		srg0200	Self-regulation: choose to pursue different goals					
		srg0400	Self-regulation: doing bad things, if they are fun					
		srg0500	Self-regulation: pleasure keeps me from getting work done					
		srg0600	Self-regulation: desire for more self-discipline					
Bullying exposure	bul0100	Bullying: frequency of teasing recently	2/3/5		Mean of listed items; higher = stronger bullying exposure			
	bul0200	Bullying: frequency of rumors recently						
	bul0300	Bullying: frequency of exclusion recently						

Note. CD = completely disagree; CA = completely agree; NC = not correct at all; FC = fully correct; Rev. = reverse-coded. Wave numbers refer to TwinLife face-to-face waves (e.g., 2 = F2F2). Scale endpoints and waves are abbreviated to reduce table width. Where cells are merged, the information applies to all items in the respective construct. Multi-item indices are calculated only when all retained items are available. Only items retained in the final indices are shown. Item-level reliability diagnostics led to the exclusion of one candidate item from parental expectations (inv0102), household chaos (hoe0200), self-regulation (srg0300), and bullying exposure (bul0400).

Table B2. Reliability of the multi-item indices

Domain	Construct	k	N	α	ω	r
Stress	Perceived school-related stress	7	851	0.77	0.77	0.33
Family	Parental expectations	2	847	-	-	0.59
	Household chaos	2	799	-	-	0.48
	Parental autonomy support	3	805	0.72	0.72	0.46
	Parental sanctions	3	843	0.78	0.78	0.54
	Academic self-concept: German	3	851	0.82	0.83	0.61
Personal	Academic self-concept: mathematics	3	851	0.89	0.89	0.73
	Intrinsic motivation: German	3	851	0.91	0.91	0.76
	Intrinsic motivation: mathematics	3	851	0.92	0.93	0.80
	Learning motivation	3	851	0.74	0.74	0.49
	Self-efficacy	3	728	0.76	0.76	0.51
School	Self-regulation	5	695	0.74	0.74	0.36
	School climate	2	846	-	-	0.36
	Teacher-student interaction	3	843	0.69	0.69	0.42
Social	Bullying exposure	3	807	0.68	0.68	0.42

Note. k = number of items; α = Cronbach's alpha; ω = McDonald's omega; r = Average inter-item r . Reliability estimates are based on unique respondents included in at least one of the German or mathematics samples of the main three-wave analysis. For two-item indices, the inter-item correlation is reported as the primary reliability estimate, while α and ω are omitted. N refers to respondents with complete information on all items of the respective index.

Table B3. Reliability of the school-related stress index by school track

Group	N	Cronbach alpha	Standardized alpha	McDonald omega total	Average inter-item correlation
All retained tracks	1786	0.77	0.77	0.77	0.32
Academic track	1020	0.78	0.78	0.78	0.33
Non-academic track	400	0.77	0.77	0.77	0.32
Comprehensive school	366	0.75	0.75	0.75	0.30

Note. Estimates are based on complete item-level cases from F2F2 among students in the three retained school-track groups. Reliability estimates are reported separately by school track as a descriptive check of internal consistency. These comparisons do not constitute a test of measurement invariance.

Table B4. Exploratory factor analysis fit of the school-related stress items

	N	No. of factors	RMSR	TLI	RMSEA	BIC	Factor correlation
One-factor solution	1930	1	0.05	0.89	0.09	109.1	-
Two-factor solution	1930	2	0.03	0.95	0.06	1.1	0.70

Note. One- and two-factor exploratory factor solutions are compared for the seven school-related stress items using complete item-level cases from F2F2 and polychoric correlations. Both solutions use minimum residual extraction; the two-factor solution uses oblimin rotation. RMSR = root mean square residual; TLI = Tucker-Lewis Index; RMSEA = root mean square error of approximation; BIC = Bayesian information criterion. Lower RMSR, RMSEA, and BIC values and higher TLI values indicate better model fit. Factor correlation is reported only for the two-factor solution. Although the two-factor solution shows better fit and the factors are strongly correlated ($r = .70$), the seven-item mean index is retained as the broad measure of perceived school-related stress.

Table B5. School-track recoding and exclusions

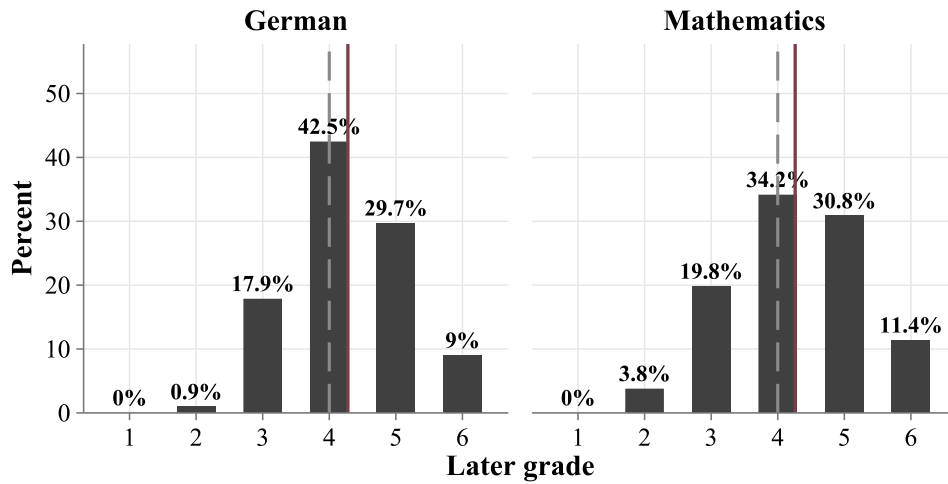
TwinLife Scale	TwinLife category	Recoded category	Retained tracks	N	Percent
1	Primary school	Excluded	No	0	0
2	Orientation school	Excluded		0	0
3	Lower secondary school	Non-academic track		10	1.2
4	Intermediate-level secondary school	Non-academic track	Yes	117	13.7
5	Integrated lower and intermediate-level secondary school	Non-academic track		39	4.6
6	Comprehensive school	Comprehensive school		156	18.3
7	Waldorf school	Excluded	No	0	0
8	Upper secondary school	Academic track	Yes	529	62.2
9	School for children with special needs	Excluded		0	0
10	Other school, namely	Excluded	No	0	0
11	Technical secondary school	Excluded		0	0

Note. The table shows the recoding of the original TwinLife school-type categories into the three analytical school-track groups. Lower secondary, intermediate-level secondary, and integrated lower/intermediate secondary schools are combined as the non-academic track; comprehensive schools remain separate; and upper secondary schools are assigned to the academic track. Other school types are excluded because they cannot be assigned unambiguously to these three analytical groups.

Appendix C. Sample and Model Diagnostics

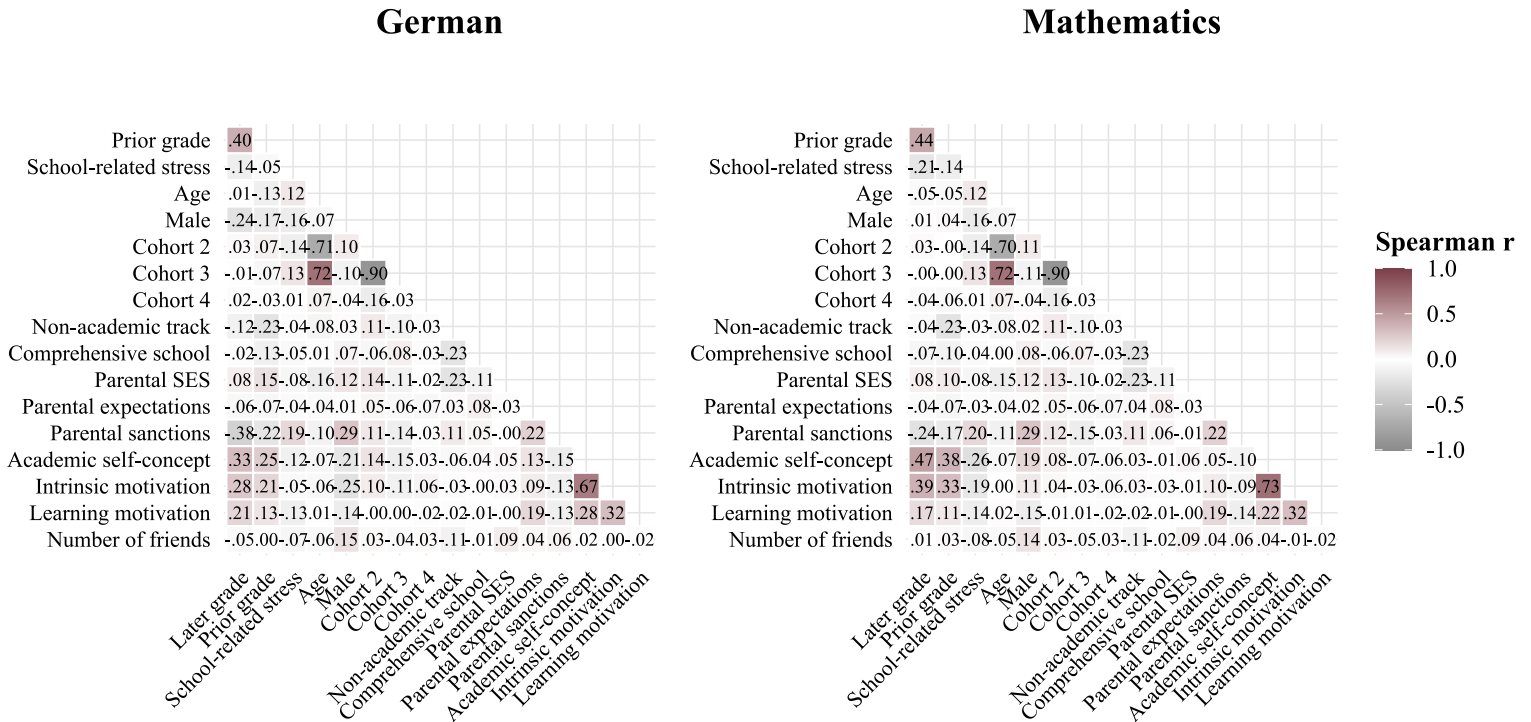
Appendix C provides descriptive checks and model diagnostics for the main three-wave LDV analyses. [Figures C1](#) and [C2](#) show the distribution of later grades and bivariate associations among the main analytical variables. [Tables C1](#) to [C7](#) assess family clustering, the sensitivity of standard errors to clustering, multicollinearity, residual behavior, heteroskedasticity, influential observations, and the robustness of the main findings to ordinal-logistic estimation. The diagnostics are used to identify problems that could affect the interpretation of the linear models rather than to require perfect distributional assumptions.

Figure C1. Distribution of later grades in the main LDV samples



Note. Bars show the percentage distribution of later reverse-coded German and mathematics grades in the respective main three-wave LDV samples. Higher values indicate better achievement.

Figure C2. Spearman correlation matrix of main analytical variables



Note. Cells report Spearman rank correlations (ρ) within the German and mathematics analytical samples. Circle size represents the absolute strength of the correlation. Correlations are estimated using pairwise complete observations; pairwise sample sizes therefore vary across variable combinations.

Table C1. Family clustering diagnostics and intraclass correlations

	N	Families	Mean cluster size	Singleton families (%)	Pair families (%)	Larger families (%)	ICC later grade	ICC final-model residual
German	795	461	1.72	36.2	55.1	8.7	0.462	0.333
Mathematics	793	458	1.73	35.8	55.2	9	0.452	0.316

Note. Cluster size refers to the number of students from the same family represented in the respective M4 estimation sample. Singleton, pair, and larger-family columns report the percentage of families represented by one, two, or three students, respectively. ICC = intraclass correlation coefficient. ICC later grade describes unconditional family-level clustering in later achievement, whereas ICC M4 residual describes the remaining family-level clustering in residuals from the final M4 model.

Table C2. Conventional and family-clustered standard errors for key model terms

Model term		b	SE conventional	SE clustered	SE ratio	p conventional	p clustered
German	Prior grade	0.319	0.037	0.04	1.07	< .001	< .001
	School-related stress	-0.158	0.066	0.067	1.01	0.017	0.018
	Stress × comprehensive school	0.073	0.139	0.146	1.05	0.599	0.617
	Stress × non-academic track	0.142	0.132	0.121	0.92	0.283	0.243
Mathematics	Prior grade	0.311	0.037	0.039	1.04	< .001	< .001
	School-related stress	-0.137	0.076	0.075	0.99	0.07	0.066
	Stress × comprehensive school	0.127	0.156	0.162	1.04	0.413	0.432
	Stress × non-academic track	-0.118	0.148	0.154	1.04	0.425	0.442

Note. Estimates are from the final M4 models. The coefficient estimates are identical across both calculations; only the standard errors and corresponding *p* values differ. SE ratio is calculated as the family-clustered standard error divided by the conventional standard error. Family-clustered standard errors are used for inference throughout the main analyses.

Table C3. Multicollinearity diagnostics for the M4 models

Predictor block		GVIF	df	Adjusted GVIF
German	School-related stress	1.54	5	1.04
	School track	1.54	5	1.04
	Prior grade	1.28	1	1.13
	Age	3.33	1	1.82
	Sex	1.28	1	1.13
	Cohort	3.34	3	1.22
	Parental SES	1.20	1	1.09
	Parental expectations	1.15	1	1.07
	Parental sanctions	1.33	1	1.15
	Academic self-concept	1.98	1	1.41
	Intrinsic motivation	1.92	1	1.39
	Learning motivation	1.20	1	1.09
	Number of friends	1.04	1	1.02
	School-related stress	1.58	5	1.05
	School track	1.58	5	1.05
Mathematics	Prior grade	1.35	1	1.16
	Age	3.22	1	1.79
	Sex	1.28	1	1.13
	Cohort	3.21	3	1.21
	Parental SES	1.18	1	1.09
	Parental expectations	1.14	1	1.07
	Parental sanctions	1.33	1	1.15
	Academic self-concept	2.50	1	1.58
	Intrinsic motivation	2.39	1	1.54
	Learning motivation	1.21	1	1.10
	Number of friends	1.04	1	1.02

Note. GVIF = generalized variance inflation factor. Diagnostics are based on the final M4 models and use predictor-level GVIFs to account for interaction terms. For predictors with more than one degree of freedom, the adjusted value $GVIF^{1/(2 \times df)}$ is reported to permit comparison across predictors. The adjusted values show no indication of problematic multicollinearity.

Table C4. Residual and heteroskedasticity diagnostics for the M4 models

	N	Mean residual	SD residual	Min residual	Max residual	Mean standardized residual	SD standardized residual	Std. residual > 2 (%)	Std. residual > 3 (%)	BP χ^2	df	BP p
German	795	0	0.73	-2.06	2.43	0	1	4.91	0.38	19.57	18	0.36
Mathematics	793	0	0.83	-2.23	2.51	0	1.00	3.53	0.13	37.68	18	< .001

Note. Residuals are from the final M4 models. Standardized residual columns report the percentage of observations with absolute standardized residuals above 2 and 3. BP = Breusch-Pagan test for heteroskedasticity. The test provides no evidence of heteroskedasticity for German but indicates heteroskedasticity for mathematics, reinforcing the use of robust family-clustered standard errors for inference.

Table C5. Influence diagnostics for the final M4 models

	N	Parameters	Cook threshold	Leverage threshold	Max Cook's D	Max leverage	Max standar. residual	N Cook > threshold	N leverage > threshold	N std. residual > 3	N any flag
German	795	19	0.005	0.048	0.033	0.268	3.299	44	46	3	76
Mathematics	793	19	0.005	0.048	0.057	0.265	3.027	47	45	1	74

Note. Potentially influential observations are identified using Cook's distance greater than $4/N$, leverage greater than $2p/N$, or an absolute standardized residual greater than 3, where p denotes the number of estimated model parameters. These criteria are diagnostic thresholds and do not imply that flagged observations should automatically be excluded.

Table C6. Sensitivity models excluding influential observations

	Term	Original b	Original SE	Original p	b without flagged cases	SE without flagged cases	p without flagged cases	N original	N restricted
German	School-related stress	-0.16	0.07	0.02	-0.13	0.06	0.03	795	751
	Stress $\times$ comprehensive school	0.07	0.15	0.62	0.07	0.12	0.55	795	751
	Stress $\times$ non-academic track	0.14	0.12	0.24	0.05	0.11	0.67	795	751
Mathematics	School-related stress	-0.14	0.07	0.07	-0.15	0.07	0.04	793	746
	Stress $\times$ comprehensive school	0.13	0.16	0.43	0.11	0.15	0.47	793	746
	Stress $\times$ non-academic track	-0.12	0.15	0.44	-0.05	0.13	0.68	793	746

Note. Restricted models exclude observations with Cook's distance greater than $4/N$ or an absolute standardized residual greater than 3; observations flagged only for high leverage are retained. The comparison focuses on the school-related stress coefficient and the two stress $\times$ school-track interaction terms. The substantive conclusions remain similar after excluding these observations.

Table C7. Ordinal-logistic robustness checks for the main achievement models

	Model	Term	OLS b	OLS SE	OLS p	Ordinal-logit b	Ordinal-logit SE	Odds ratio	Ordinal-logit p
German	M3	School-related stress	-0.3	0.07	< .001	-0.71	0.17	0.49	< .001
		Stress $\times$ comprehensive school	0.06	0.16	0.71	0.09	0.41	1.1	0.82
		Stress $\times$ non-academic track	0.12	0.13	0.36	0.22	0.32	1.25	0.49
	M4	School-related stress	-0.16	0.07	0.02	-0.41	0.17	0.67	0.02
		Stress $\times$ comprehensive school	0.07	0.15	0.62	0.18	0.4	1.2	0.65
		Stress $\times$ non-academic track	0.14	0.12	0.24	0.29	0.31	1.34	0.35
Mathematics	M3	School-related stress	-0.26	0.08	0.001	-0.55	0.17	0.58	0.002
		Stress $\times$ comprehensive school	-0.02	0.18	0.91	-0.01	0.41	0.99	0.99
		Stress $\times$ non-academic track	-0.2	0.15	0.18	-0.35	0.31	0.71	0.26
	M4	School-related stress	-0.14	0.07	0.07	-0.3	0.18	0.74	0.09
		Stress $\times$ comprehensive school	0.13	0.16	0.43	0.42	0.39	1.52	0.29
		Stress $\times$ non-academic track	-0.12	0.15	0.44	-0.17	0.35	0.84	0.62

Note. OLS and ordinal-logit coefficients are not directly comparable in magnitude. For the ordinal-logit models, odds ratios below 1 indicate lower odds of being in a higher achievement category. The robustness assessment therefore focuses on the direction, statistical uncertainty, and substantive interpretation of the key terms rather than on coefficient magnitude. The ordinal-logit models lead to the same substantive conclusions as the corresponding OLS models.

Figure C3. Q-Q plots and residual distributions for the final M4 models

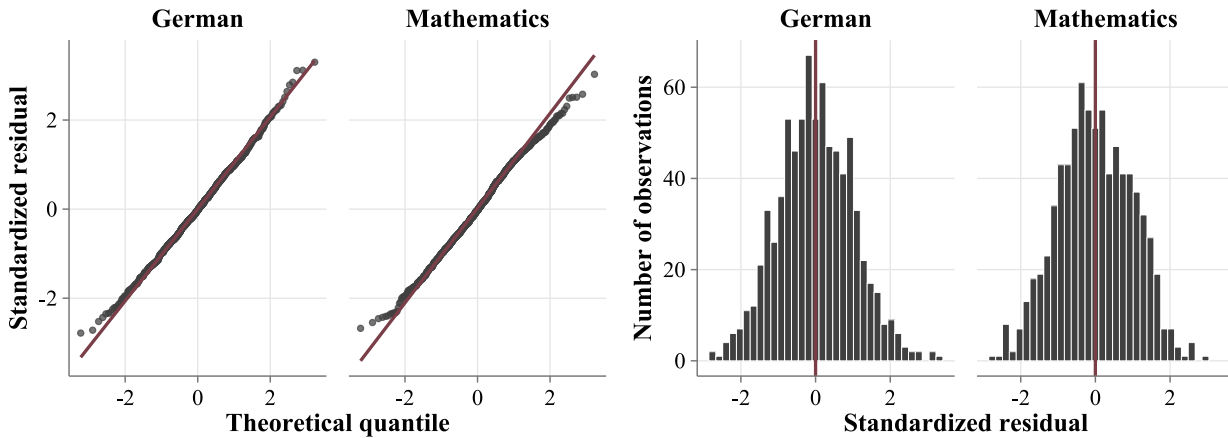


Figure C4. Residuals versus fitted values and scale-location plots for the final M4 models

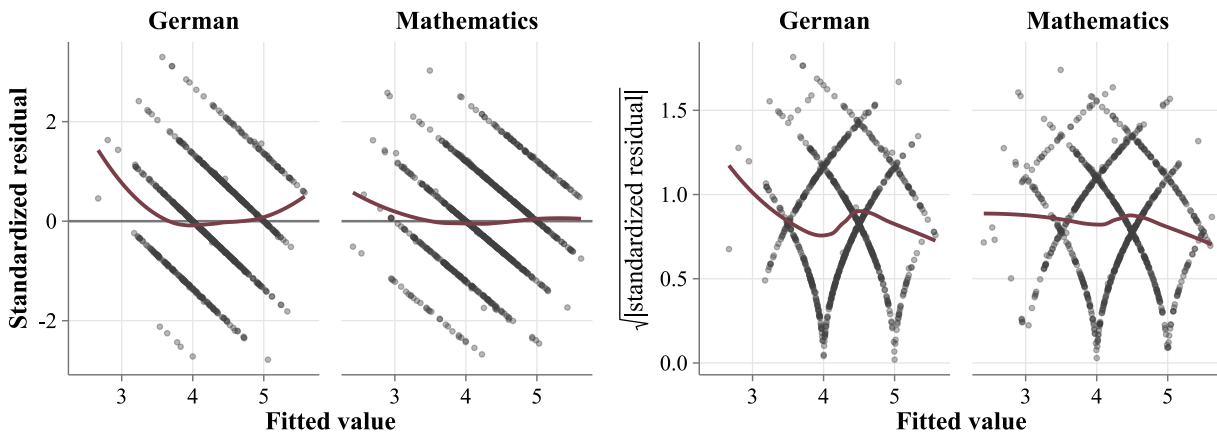
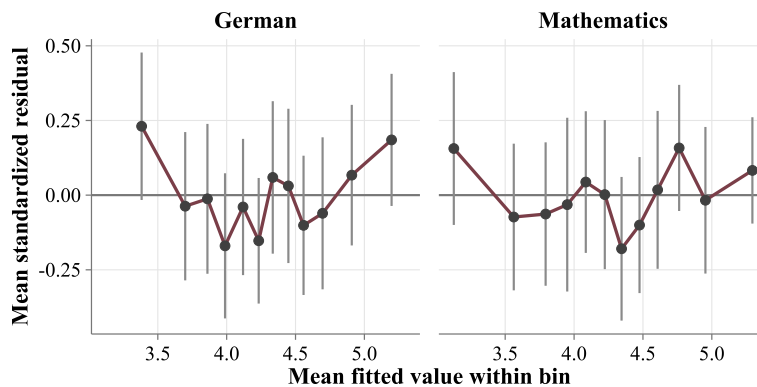


Figure C5. Binned residual plots for the final M4 models

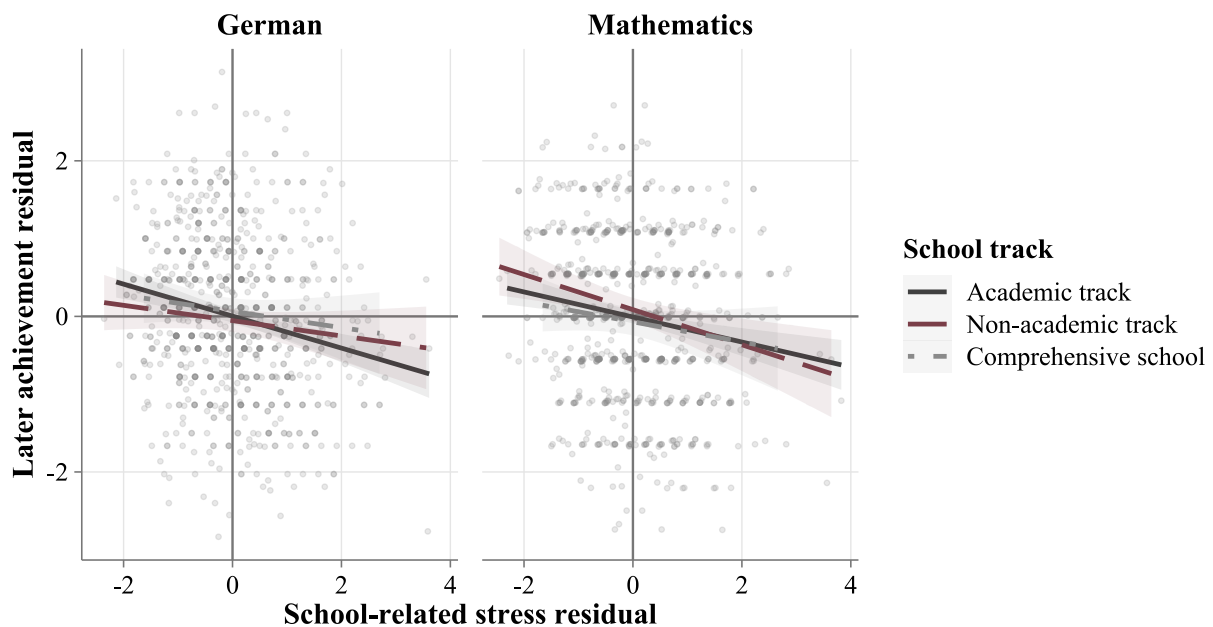


Shared Note. Diagnostic plots are based on the final M4 models for German and mathematics. Q-Q plots and residual histograms assess the distribution of standardized residuals. Residual-versus-fitted plots are used to detect systematic patterns in the residuals, while scale-location plots assess whether residual dispersion changes across fitted values. Binned residual plots show mean standardized residuals across fitted-value bins and provide an additional check for systematic deviations in the conditional mean. The diagnostics are used to identify substantial model misspecification rather than to require exact normality or constant variance.

Appendix D. Moderation Diagnostics and Slope Contrasts

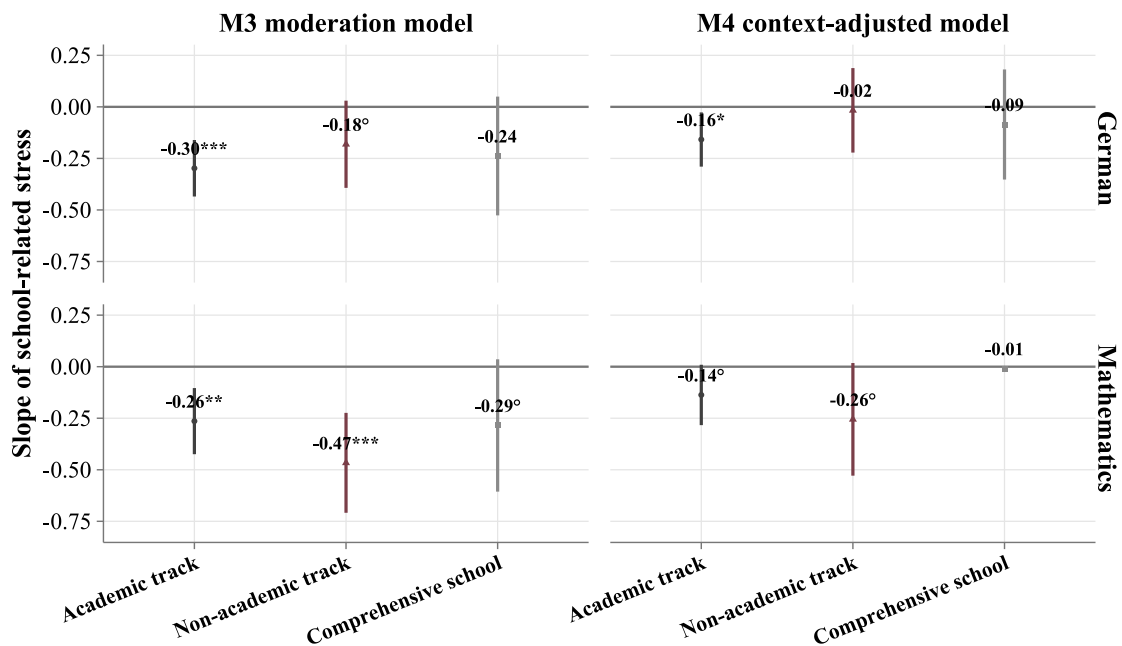
Appendix D provides supplementary diagnostics for the school-track moderation analyses. [Figure D1](#) visualizes the adjusted stress-achievement pattern after accounting for prior achievement and the basic controls. [Figures D2](#) and [D3](#) translate the M3 moderation and M4 context-adjusted models into track-specific simple slopes and direct pairwise slope contrasts. [Table D1](#) complements these estimates with joint tests of the school-related stress $\times$ school-track interaction terms. Together, these analyses assess whether the stress-achievement association differs systematically across school tracks.

Figure D1. Residualized association between school-related stress and later achievement by school track



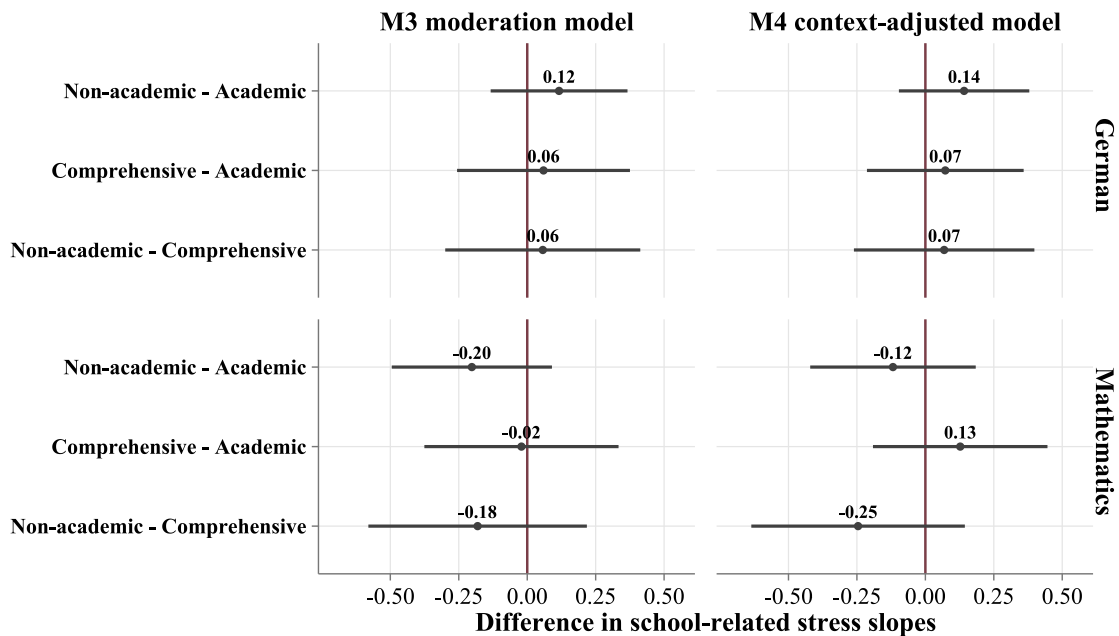
Note. School-related stress and later achievement are each residualized for prior subject-specific achievement, age, sex, and cohort and subsequently standardized. Points show the resulting residualized observations. Lines show descriptive linear associations within each school track; shaded areas indicate 95% confidence intervals around these trends. The figure is a supplementary visualization and does not replace the formal moderation tests.

Figure D2. Track-specific simple slopes of school-related stress



Note. Points show track-specific school-related stress slopes from the M3 moderation and M4 context-adjusted models for German and mathematics. Intervals indicate 95% confidence intervals based on family-clustered standard errors. Negative slopes indicate lower later achievement at higher levels of school-related stress. ° $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

Figure D3. Pairwise contrasts of school-related stress slopes across school tracks



Note. Points show pairwise differences between track-specific school-related stress slopes from the M3 moderation and M4 context-adjusted models. Contrasts are calculated as the first named school track minus the second. Intervals indicate 95% confidence intervals based on family-clustered standard errors. Confidence intervals including zero provide no evidence of a difference between the corresponding slopes. ° $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

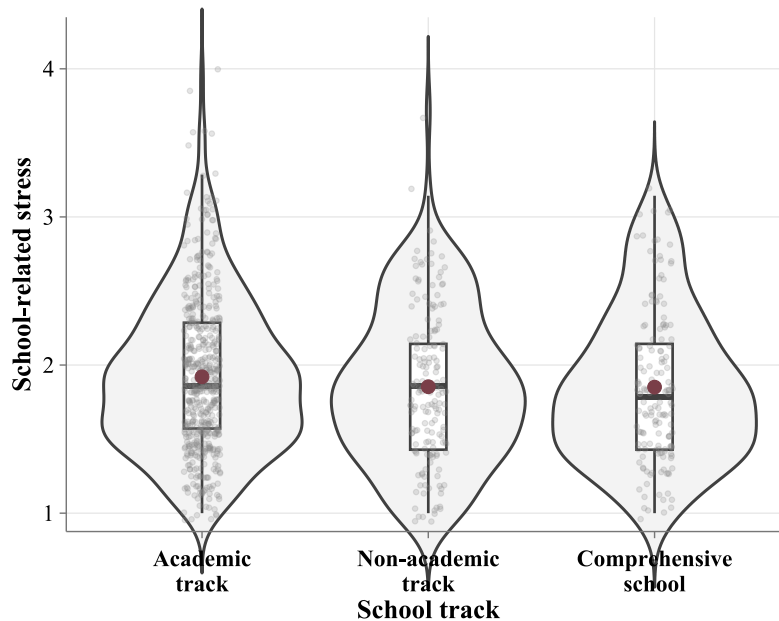
Table D1. Joint Wald tests of school-related stress $\times$ school-track interactions

	Model	N	Families	Interaction terms tested	Wald χ^2	df	p-value
German	M3 moderation model	795	461	2	0.87	2	0.65
	M4 context-adjusted model	795	461	2	1.43	2	0.49
Mathematics	M3 moderation model	793	458	2	1.90	2	0.39
	M4 context-adjusted model	793	458	2	1.53	2	0.46

Note. Joint Wald tests assess whether the two school-related stress $\times$ school-track interaction terms are jointly equal to zero. Tests are based on family-clustered variance-covariance matrices. Non-significant tests provide no evidence that the school-related stress-achievement association differs systematically across the three school tracks.

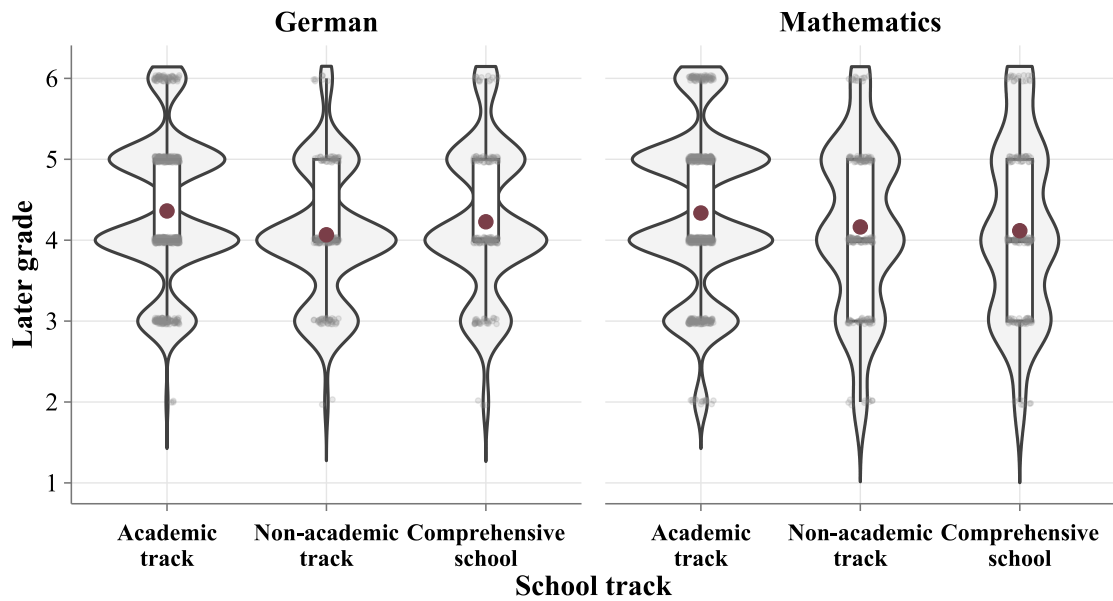
Appendix E. Descriptive Distributions by School Track

Appendix E presents the distributions of perceived school-related stress and later achievement across the three school-track groups. The figures complement the descriptive comparisons reported in [Section 4.1](#) and are unadjusted. They are not intended as tests of school-track moderation.

Figure E1. Distribution of school-related stress by school track

Note. Violin plots show the distribution of perceived school-related stress across school tracks. Boxplots indicate medians and interquartile ranges, red dots indicate means, and points represent individual observations. School-related stress ranges from 1 to 4. Because the measure is bounded, the smoothed violin densities should be interpreted as descriptive approximations. The figure is unadjusted.

Figure E2. Distribution of later German and mathematics grades by school track



Note. Violin plots show the distribution of later reverse-coded grades by school track. Boxplots indicate medians and interquartile ranges; red dots indicate means. Points show individual observations. Higher values indicate better achievement. Because grades are discrete ordinal categories, the smoothed violin density is only a visual approximation; the observed points and boxplots show the actual distributional information more directly. The figure is descriptive and unadjusted.

Appendix F. Supplementary Two-Wave Analyses

Appendix F reports supplementary two-wave models relating school-related stress at T1 (F2F2) to German and mathematics grades at T2 (F2F3). Unlike the main three-wave LDV models, these specifications do not adjust for prior achievement at T0 and therefore provide a less stringent supplementary assessment of the stress-achievement association. Models are estimated separately for German and mathematics with family-clustered standard errors. Within each subject, M1 to M4 are estimated on the same complete-case sample, allowing coefficient changes across specifications to be compared without changes in sample composition. The model sequence estimates the basic stress-achievement association, adds school track, tests the stress $\times$ school-track interaction, and finally adds the selected contextual covariates. Joint Wald tests assess the interaction terms simultaneously.

Table F1. Supplementary two-wave sample composition by school track

		N	Families	Stress		Grades	
				M	(SD)	M	(SD)
German	Overall	548	310	1.84	0.49	4.3	0.89
	Academic track	348	204	1.86	0.48	4.35	0.91
	Non-academic track	99	66	1.85	0.53	4.12	0.8
	Comprehensive school	101	61	1.78	0.5	4.28	0.85
Mathematics	Overall	544	310	1.84	0.49	4.33	1.01
	Academic track	347	204	1.85	0.48	4.38	0.97
	Non-academic track	96	65	1.85	0.53	4.3	1.01
	Comprehensive school	101	61	1.79	0.5	4.2	1.13

Note. The table describes the complete-case samples used throughout Tables F2 to F4. Sample inclusion requires complete information for the variables defining the supplementary two-wave analysis and the final M4 contextual specification. School-related stress is measured at T1 (F2F2) and later grades at T2 (F2F3). Means are reported by outcome and school track. Grades are reverse-coded so that higher values indicate better achievement. *M* = mean; *SD* = standard deviation.

Table F2. Supplementary two-wave models predicting later German grades

Outcome: German grades								
Variable	M1 basic		M2 track		M3 moderation		M4 context	
	<i>b</i>	(SE)	<i>b</i>	(SE)	<i>b</i>	(SE)	<i>b</i>	(SE)
Intercept	4.58***	(0.32)	4.61***	(0.33)	4.62***	(0.32)	3.93***	(0.43)
School-related stress	-0.26***	(0.08)	-0.26***	(0.08)	-0.31**	(0.10)	-0.11	(0.09)
Age	0.01	(0.05)	0.01	(0.05)	0.01	(0.05)	-0.001	(0.05)
Male	-0.52***	(0.09)	-0.53***	(0.08)	-0.53***	(0.08)	-0.31***	(0.09)
Cohort 2	0.02	(0.31)	0.04	(0.32)	0.03	(0.32)	-0.001	(0.27)
Cohort 3	-0.21	(0.44)	-0.16	(0.44)	-0.19	(0.44)	-0.19	(0.43)
Cohort 4	0.04	(0.44)	0.01	(0.44)	0.02	(0.43)	-0.09	(0.36)
School track (ref.: Academic track)								
Non-academic track			-0.25*	(0.11)	-0.25*	(0.11)	-0.10	(0.11)
Comprehensive school			-0.04	(0.11)	-0.04	(0.11)	0.04	(0.11)
Interaction (ref.: Academic track)								
Stress × non-academic track					0.14	(0.19)	0.18	(0.17)
Stress × comprehensive school					0.10	(0.18)	0.14	(0.17)
Context covariates								
Parental SES							0.006*	(0.003)
Parental expectations							-0.04	(0.05)
Parental sanctions							-0.22***	(0.04)
Academic self-concept							0.22**	(0.07)
Intrinsic motivation							0.02	(0.06)
Learning motivation							0.12°	(0.06)
Number of friends							-0.01	(0.01)
N	548		548		548		548	
Families	310		310		310		310	
R²	0.10		0.11		0.11		0.23	
Adjusted R²	0.09		0.10		0.10		0.21	

Note. Entries are unstandardized OLS coefficients (*b*) with family-clustered standard errors in parentheses. All models are estimated on the same complete-case sample. Grades are reverse-coded so that higher values indicate better achievement. Female, Cohort 1, and the academic track are the reference categories. School-related stress and age are mean-centered; parental SES is additionally mean-centered in M4. ° $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$. Parental SES and coefficients that would otherwise round to 0.00 are reported with three decimal places.

Table F3. Supplementary two-wave models predicting later mathematics grades

Outcome: Mathematics grades								
Variable	M1 basic		M2 track		M3 moderation		M4 context	
	b	(SE)	b	(SE)	b	(SE)	b	(SE)
Intercept	4.27***	(0.26)	4.29***	(0.26)	4.27***	(0.26)	2.96***	(0.41)
School-related stress	-0.43***	(0.10)	-0.43***	(0.10)	-0.40**	(0.13)	-0.18°	(0.10)
Age	-0.04	(0.07)	-0.04	(0.07)	-0.04	(0.07)	-0.03	(0.06)
Male	-0.09	(0.10)	-0.08	(0.10)	-0.08	(0.10)	-0.16	(0.10)
Cohort 2	0.11	(0.26)	0.14	(0.26)	0.16	(0.26)	0.18	(0.20)
Cohort 3	0.21	(0.54)	0.26	(0.56)	0.31	(0.56)	0.31	(0.43)
Cohort 4	-0.004	(0.31)	-0.02	(0.31)	-0.02	(0.31)	0.60°	(0.36)
School track (ref.: Academic track)								
Non-academic track			-0.09	(0.12)	-0.09	(0.12)	0.06	(0.12)
Comprehensive school			-0.20	(0.15)	-0.20	(0.15)	0.01	(0.14)
Interaction (ref.: Academic track)								
Stress × non-academic track					-0.24	(0.22)	-0.07	(0.20)
Stress × comprehensive school					0.07	(0.26)	0.30	(0.22)
Context covariates								
Parental SES							0.008**	(0.003)
Parental expectations							0.000	(0.05)
Parental sanctions							-0.17***	(0.05)
Academic self-concept							0.41***	(0.07)
Intrinsic motivation							0.08	(0.06)
Learning motivation							-0.01	(0.06)
Number of friends							-0.007°	(0.005)
N	544		544		544		544	
Families	310		310		310		310	
R²	0.05		0.05		0.06		0.27	
Adjusted R²	0.04		0.04		0.04		0.25	

Note. Entries are unstandardized OLS coefficients (*b*) with family-clustered standard errors in parentheses. All models are estimated on the same complete-case sample. Grades are reverse-coded so that higher values indicate better achievement. Female, Cohort 1, and the academic track are the reference categories. School-related stress and age are mean-centered; parental SES is additionally mean-centered in M4. ° $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$. Parental SES and coefficients that would otherwise round to 0.00 are reported with three decimal places.

Table F4. Joint Wald tests of school-track moderation in the two-wave models

	Model	N	Families	χ^2	df	p
German	M3 moderation	548	310	0.68	2	0.71
	M4 context-adjusted			1.47		0.48
Mathematics	M3 moderation	544		1.57		0.46
	M4 context-adjusted			2.17		0.34

Note. Joint Wald tests assess whether the two school-related stress × school-track interaction terms are simultaneously equal to zero. Tests are based on family-clustered variance-covariance matrices. The academic track is the reference category, with the two interaction terms representing deviations for the non-academic track and comprehensive schools. M3 refers to the moderation model without contextual covariates; M4 refers to the corresponding context-adjusted specification reported in Tables F2 and F3.

Appendix G. Supplementary Cross-Sectional Analyses

Appendix G provides a cross-sectional diagnostic analysis using school-related stress and achievement measured within F2F5 (T4). Because stress and achievement are measured at the same wave, these models do not establish temporal ordering and are not interpreted as prospective evidence. Their purpose is to assess whether the negative stress-achievement association and the school-track moderation pattern observed in the main analyses are also visible in a later analytical sample. Models are estimated separately for German and mathematics with family-clustered standard errors. M1 includes school-related stress and the basic controls, M2 adds school track, and M3 adds the school-related stress $\times$ school-track interaction.

Table G1. Later-wave cross-sectional sample composition by school track

	School track	N	Families	Mean stress	SD stress	Mean grade	SD grade
German	Academic track	361	207	2.00	0.52	4.54	0.87
	Non-academic track	73	47	2.01	0.43	4.30	0.84
	Comprehensive school	97	60	2.01	0.56	4.39	0.94
Mathematics	Academic track	361	207	2.00	0.52	4.42	1.07
	Non-academic track	72	47	2.02	0.42	4.31	0.99
	Comprehensive school	97	60	2.01	0.56	4.32	1.05

Note. The table describes the complete-case samples used in the later-wave cross-sectional models. School-related stress and achievement are both measured at F2F5. *N* denotes the number of students and Families the number of unique families represented within each school-track group. *M* = mean; *SD* = standard deviation. Grades are reverse-coded so that higher values indicate better achievement.

Table G2. Later-wave cross-sectional models predicting German grades

Outcome: German grades						
Variable	M1 basic		M2 track		M3 moderation	
	<i>b</i>	(SE)	<i>b</i>	(SE)	<i>b</i>	(SE)
Intercept	4.62***	(0.10)	4.69***	(0.11)	4.69***	(0.11)
School-related stress	-0.20*	(0.09)	-0.20*	(0.09)	-0.22*	(0.10)
Age	-0.04	(0.03)	-0.05	(0.03)	-0.05	(0.03)
Male	-0.33***	(0.09)	-0.34***	(0.09)	-0.34***	(0.09)
Cohort 2	0.09	(0.20)	0.10	(0.20)	0.09	(0.20)
School track (ref.: Academic track)						
Non-academic track			-0.32*	(0.14)	-0.33*	(0.14)
Comprehensive school			-0.12	(0.12)	-0.12	(0.12)
Stress $\times$ school track (ref.: Academic track)						
Stress $\times$ non-academic track					-0.03	(0.27)
Stress $\times$ comprehensive school					0.16	(0.21)
N	531		531		531	
Families	289		289		289	
R²	0.05		0.07		0.07	
Adjusted R²	0.04		0.06		0.05	

Note. Entries are unstandardized OLS coefficients (*b*) with family-clustered standard errors in parentheses. All models are estimated on the same complete-case sample. Grades are reverse-coded so that higher values indicate better achievement. Female, Cohort 1, and the academic track are the reference categories. School-related stress and age are mean-centered. ° $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

Table G3. Later-wave cross-sectional models predicting mathematics grades

Outcome: Mathematics grades						
Variable	M1 basic		M2 track		M3 moderation	
	b	(SE)	b	(SE)	b	(SE)
Intercept	4.47***	(0.12)	4.52***	(0.12)	4.51***	(0.13)
School-related stress	-0.37***	(0.10)	-0.37***	(0.11)	-0.40**	(0.12)
Age	-0.04	(0.04)	-0.05	(0.04)	-0.05	(0.04)
Male	-0.14	(0.11)	-0.15	(0.11)	-0.16	(0.11)
Cohort 2	-0.05	(0.25)	-0.05	(0.25)	-0.03	(0.26)
School track (ref.: Academic track)						
Non-academic track			-0.21	(0.16)	-0.22	(0.15)
Comprehensive school			-0.06	(0.13)	-0.06	(0.13)
Stress × school track (ref.: Academic track)						
Stress × non-academic track					0.71°	(0.37)
Stress × comprehensive school					-0.17	(0.21)
N	530		530		530	
Families	289		289		289	
R²	0.06		0.06		0.08	
Adjusted R²	0.05		0.05		0.06	

Note. Entries are unstandardized OLS coefficients (*b*) with family-clustered standard errors in parentheses. All models are estimated on the same complete-case sample. Grades are reverse-coded so that higher values indicate better achievement. Female, Cohort 1, and the academic track are the reference categories. School-related stress and age are mean-centered. ° $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

Table G4. Joint Wald tests of school-track moderation in the later-wave cross-sectional models

	Model	N	Families	No. of interaction terms	χ^2 (chisq)	df	p
German	M3 moderation	531	289	2	0.58	2	0.75
Mathematics	M3 moderation	530			5.07		0.08

Note. Joint Wald tests assess whether the two school-related stress × school-track interaction terms are simultaneously equal to zero. Tests are based on family-clustered variance-covariance matrices. The academic track is the reference category, with the interaction terms representing deviations for the non-academic track and comprehensive schools.

Appendix H. Exploratory Prospective RQ4 Models

Appendix H provides supplementary material for RQ4, which examines whether contextual conditions at T1 (F2F2) are associated with school-related stress at T2 (F2F3) after accounting for school-related stress at T1, and whether these associations differ across school tracks. Models are estimated separately for the school, family, social, and personal contexts using family-clustered standard errors. Because numerous contextual coefficients and interaction terms are examined, the analyses are exploratory and individual coefficient-level findings are interpreted cautiously.

Table H1. Availability of RQ4 model variables across waves

Construct		F2F1 N	F2F2 N	F2F3 N	Used wave(s)
Controls	School-related stress	0	1930	1498	F2F2 and F2F3
	School track	3745	1972	1555	
	Age	10298	6499	5272	
	Cohort	10298	6500	5274	
	Sex	10298	6500	5274	
Context					
School	School climate	0	1935	1499	
	Teacher-student interaction	0	1930	1487	
Family	Household chaos	6331	2921	0	F2F2
	Parental SES	8881	5717	4551	
	Parental autonomy support	3958	2088	0	
	Parental expectations	4367	2198	0	
	Parental sanctions	4370	2190	0	
Social	Bullying exposure	0	2076	1502	
	Loneliness	7466	4217	4388	
	Number of friends	6860	4515	4939	
Personal	Academic self-concept	4872	3765	1342	
	Intrinsic motivation	4909	3811	3010	
	Learning motivation	4459	2221	1650	
	Self-efficacy	7497	3938	3214	
	Self-regulation	0	3779	0	

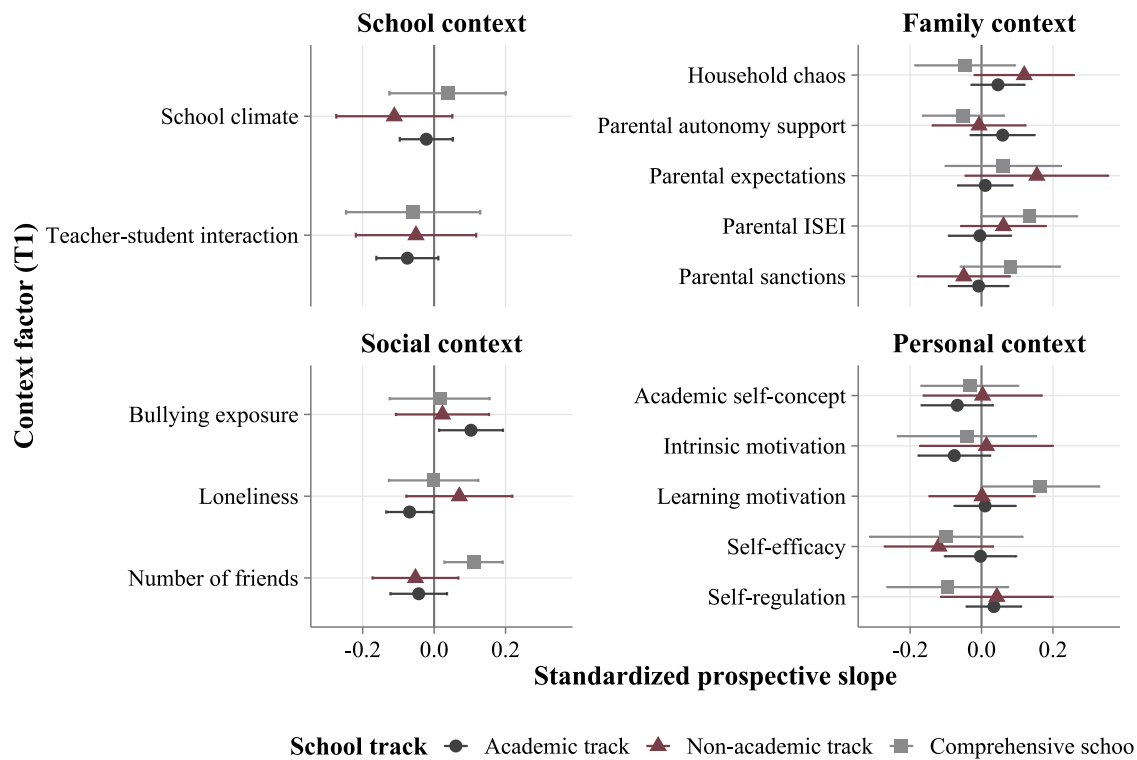
Note. The table reports variable-level numbers of valid observations across F2F1 to F2F3. In the RQ4 analyses, school-related stress at F2F2 is used as the prior-stress control and school-related stress at F2F3 as the outcome; school track, controls, and contextual predictors are measured at F2F2. F2F1 is shown only to document variable availability across the earlier waves. Counts represent variable-level availability and therefore do not correspond to the complete-case estimation samples of the context-specific models.

Table H2. Joint Wald tests of context-by-school-track interactions in prospective models of later school-related stress

	N	Families	No. of interaction terms	χ^2	df	p
School context	1129	579	4	4.41	4	0.66
Family context	973	521	10	11.42	10	0.33
Social context	1039	560	6	12.29	6	0.06
Personal context	798	500	10	11.35	10	0.33

Note. Joint Wald tests assess whether all context $\times$ school-track interaction terms within each context block are simultaneously equal to zero. Tests are based on family-clustered variance-covariance matrices. *N* denotes the number of students and Families the number of unique family clusters in the respective complete-case model sample. Non-significant tests provide no evidence that the contextual associations differ systematically across school tracks.

Figure H1. Complete set of school-track-specific contextual associations with later school-related stress



Note. Points show standardized school-track-specific slopes from the context $\times$ school-track models; intervals indicate 95% confidence intervals based on family-clustered standard errors. Contextual predictors are measured at F2F2 and later school-related stress at F2F3. Models control for school-related stress at F2F2, age, sex, cohort, and school track. Both stress measures, age, and all continuous contextual predictors are standardized. Each context block is estimated on its respective complete-case sample. Positive estimates indicate higher later stress at higher levels of the contextual predictor, conditional on the remaining model covariates.

Declaration of Independent Work

I hereby declare that I have written the present thesis independently and have not used any sources other than those indicated. All passages taken from other works, either verbatim or in substance, have been clearly identified and referenced accordingly.

Krefeld, Germany

August 20, 2026

Richard Höchter

Disclosure Statement on the Use of AI Tools

	Form of use	AI Tool (version)	Notes
	To-do lists for the process of writing papers and individual work steps		
X	Research question, social science relevance, and research design	OpenAI ChatGPT-5.6 Sol	Critical reflection of research questions, (longitudinal) design, model specifications, and robustness/appendix analyses
	Outline of the work and structure of argumentation		
X	Literature search	Consensus AI Deep Review	Support in identifying niche literature and additional relevant research directions not readily found through conventional database searches
X	Data analysis and interpretation (including generation of tables and figures)	OpenAI ChatGPT-5.6 Sol	Support with R code for statistical analyses
	Synthesis of the material used		
X	Checking spelling, syntax, and sentence structure	OpenAI ChatGPT-5.6 Sol	Proofreading for spelling, grammar, syntax, and sentence structure
X	Language style in terms of semantics and phrasing	OpenAI ChatGPT-5.6 Sol	Revision of academic phrasing for clarity, concision, readability, and consistency
	Translation from a source language or into a foreign language		
X	Text generation and collaborative writing with AI support	OpenAI ChatGPT-5.6 Sol	Collaborative drafting and revision of individual passages based on the author's arguments and analyses
	Formal editing and assistance with regard to format specifications, layout requirements, and design		
	Image creation, e.g., photography, illustration, table, figure, video		